Coral tissue growth and regeneration conjoin the advent of adult stem cell-like cells

Jonathan Levanoni

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE MASTER'S DEGREE

University of Haifa

Faculty of Natural Sciences

Leon H. Charney School of Marine Sciences

Department of Marine Biology

November, 2021

Coral tissue growth and regeneration conjoin the advent of adult stem cell-like cells

By: Jonathan Levanoni

Supervised by: Prof. Baruch Rinkevich

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE MASTER'S DEGREE

University of Haifa

Faculty of Natural Sciences

Leon H. Charney School of Marine Sciences

Department of Marine Biology

November, 2021

Approved by: **Prof. Baruch Rinkevich** Date: **25 11 2021**

(Supervisor)

Approved by: **Prof. Smadar Ben Tabou de Leon** Date: **20 12 2021**

(Chairperson of Master's studies Committee)

To my family and friends.

Table of contents

List of tables V
List of figures V
List of supplementary materials VI
Table S1: Primers used for qPCR assay. VI
Abstract VII
Authors contribution VII
Introduction 1
Regeneration 1
Wound healing and regeneration 2
Stem cells 3
Article draft: 5
Summary statement 5
Abstract 5
Introduction 6
Materials and methods 9
Collection and maintenance 9
Fragmentation and lateral growth 9
Regeneration assays 10
Histology 10
Immunohistochemistry - Phosphorylated Histone H3, Piwi 10
Quantitative PCR 11
Results 12
Morphological outcomes 12
Histological observations 14
Immunohistochemistry 18
qPCR analyses 21
Discussion 23
Acknowledgements 25
competing interests 25
Funding 25
Bibliography 26
Supplementary 31

Table S1: Primers used for qPCR assay 31
Thesis bibliography 32
Supplementary 37

List of tables

Table 1. The distribution of fully regenerating assays (at 9 days post injury) in the three studied *S. pistillata* genet. 18
Table 2. Distribution of number of polyps emerged by 9dpi in fully regenerated assays for each genet. 20

List of figures

Fig. 1. Two *S. pistillata* nubbins with grown and 2D spread tissue and skeleton over the glass slide ensuing about 70 days of growth following fragmentation (margins indicated by white arrows). 6
Fig. 2. *Stylophora pistillata* regenerating nubbins. Daily images of wound areas in two regeneration assays, followed up to 9 days from onset. Both wounds were circular, each 1.5mm diameter. A-J: Fully regenerated case by day 6; K-T: Partial regeneration. Algal growth in the wound area. Yellow dashed lines indicate the wound's edge. Note the 'pale' contour of the regenerated tissue, indicating the tissue front. 18
Fig. 3. Differences between the three colonies (genet 1-3) along the regeneration process. (A) Percentage (±s.e) of regenerated surface area along nine days from injury. (B) Calculated regeneration rates for all fully regenerated injuries in each genet. Similar letters indicate non-significant differences (Mann-Whitney U-test, $p<0.05$). 19
Fig. 4. Fully regenerated tissue of *S. pistillata* at 9 dpi. (A) no polyp in the regenerating area; (B-D) 1-3 silhouettes of new polyps appeared (yellow dashed circles). 19
Fig. 5. *S. pistillata* cross sections at the injury sites or at spreading tissue edges, stained with hematoxylin & eosin. (A) newly injured tissue (0dpi); (B, F) tissue regrowing into the injury site at 2dpi; (C, G) tissue regrowing into the injury site at 5dpi, featuring epithelial epidermal epithelium expansion; (D) fully regenerated tissue at 6dpi, (E, H) a spreading nubbin tissue edge; (I) a schematic illustration of proposed tissue regeneration and growth process (from left to right): 0dpi- incision cutting through all tissues, 1-2dpi-initial closure of the injury and regeneration and 3dpi-full regeneration- addendum formation and fast regeneration. (A-E) magnification X400, bars=50µm, (F-H) magnification X1000, bars=10µm. ep-epidermis, ga-gastrodermis, me-mesoglea, zo-zooxanthellae, cm-cell mass, ca-calicoblastic layer, dashed red line indicates the cutting line in 0dpi (A) and asterisks indicate the most distal part of the calicoblastic layer. 23
Fig. 6. *S. pistillata* cross sections at the injury sites and the naturally spreading tissue edges on the substrate, stained with anti-phosphohistone H3 ($pH3^+$) antibodies (red) and DAPI (blue). A1 to D1 are negative controls. (A, A1) newly injured tissue (0dpi); (B, B1) regenerating tissue at 2dpi) (C, C1) regenerating tissue at 5dpi, featuring epidermal epithelium expansion; (D, D1) a spreading nubbin tissue edge. (A, A1) scale bars=500µm. (B-D, B1-D1) scale bars=50µm. Arrowheads depict the zooxanthellae. Asterisks indicate aggregates of $pH3^+$ cells. 25
Fig. 7. *S. pistillata* cross sections at the injury sites or spreading tissue edges stained with anti-Piwi antibodies (red) and DAPI (blue). (A, A1) newly injured tissue (0dpi); (B, B1) tissue regrowing into the injury site at 2dpi;, (C, C1) tissue regrowing into the injury site at 5dpi, featuring epidermal epithelium expansion; (D, D1) a spreading nubbin tissue edge. (A1-D1) Negative controls. Magnification X1000; scale bars=20µm; arrowheads indicate zooxanthellae and asterisks indicate $Piwi^+$ cells aggregations. 27
Fig. 8. Relative qPCR analyses for 7 stemness genes and transcription factors. Tissues from four biological states were sampled: naïve tissue (RS- Resting State), at onset of regeneration, Reg (10 hours post injury), FR (fully regenerated) and tissues collected from the naturally growing tissue edges (TE) of nubbins. The results are represented as fold change at log_2 scale. Repeated-measures ANOVA (p value indicated above each 28

examined gene) and LSD (Fisher's Least Significant Difference) post-hoc *p<0.05 **p<0.001 for each state at each gene.

List of supplementary materials

Table S1: Primers used for qPCR assay. 46

Table S2: Injury surface area [mm^2] for each nubbin at each day. 46

Table S3: Statistical analyses to total regeneration rate between each genet. 48

Figure S4: Scatter plot of initial injury size (x axis) and days to full regeneration (y axis) and Spearman's test for correlation between them. 48

Figure S5: Line plot of injury days to full regeneration (x axis) and surface area (y axis). 49

Figure S6: Chi2 test to number of polyps rebuilt following tissue regeneration. 50

Coral tissue growth and regeneration conjoin the advent of adult stem cell-like cells

Jonathan Levanoni

Abstract

Although regeneration processes in the Anthozoa have been intensively studied for more than a century, the cellular mechanisms by which tissue regeneration are executed are not understood and the putative adult stem cells (ASC) are undisclosed. Here we aimed to elucidate morphological, cellular and some gene expression patterns in injured spreading tissues of nubbins from the branching coral *Stylophora pistillata* and in the margins of fast-growing tissues. Tissue regeneration following small circular incisions (each 2.77mm^2), where a single polyp had been removed, was followed using microphotography, histological staining, pH3 and Piwi immunohistochemistry and qPCR for gene expression seven stemness markers: *Piwi1*, *Nanos-1* homologue, *Tudor5-like*, *Tudor7-like*, *Boule-like isoform*, *Sox2-like* and *Myc1-like*. Regeneration of the small injuries was completed within 9 days in 84.5% (n=64) of the assays. In nearly 35% of the cases a single polyp was regrown within this time, while in 60% no polyp regrew and in about 6% more than one polyp appeared on the regenerating margins. Results also revealed the formation of an 'addendum', an ectodermal epithelium in growing edges (in regenerating and normal growing fronts) inhabited by a cluster of small cells featuring dense nuclei, resembling ASCs. Many of these cells express pH3 as well as Piwi proteins. Most of the stemness marker genes were upregulated in newly regenerated tissue or in the spreading coral tissue. This study revealed for the first time in the Anthozoa the presence and participation of ASC-like cells in regenerating tissues and in naturally growing tissue fronts.

Authors contribution

Jonathan Levanoni – Primary author, data collection and analysis for all chapters.

Amalia Rosner – Participation and assistance in qPCR segment.

Ziva Lapidot – Participation and assistance in histology and immunohistochemistry segments.

Buki Rinkevich – Supervisor

Introduction

Scleractinian corals (Anthozoa, Cnidaria) are a group of marine organisms, mostly colonial, featuring radially symmetrical colonial units, called polyps (Ruppert et al., 2004; Veron and Stafford-Smith, 2000) who share intracolonial metabolic properties (Graham and Nash, 2013). They usually form large and massive reef formations over the centuries, through the deposition of calcium carbonate skeletons (Muscatine and Cernichiari, 1969). Thus, in the hot and oligotrophic tropical waters, the reef building corals constitute as the building blocks and the primary mainframe for reefs (Veron and Stafford-Smith, 2000). Moreover, the coral reefs provide many ecosystem services, such as coastal protection from waves (Cesar et al., 2003), fisheries, by providing a refuge for many species of fish, molluscs and crustaceans (Grandcourt and Cesar, 2003), recreational activities and tourism (Brander et al., 2007) and more.

Stylophora pistillata is a common coral in the indo-pacific reefs, which include the Gulf of Eilat (Loya, 1976). It is a branching coral, usually forming bush-like colonies growing from a single stalk to all directions, with branches, typically, grow and divide in dichotomy, which makes it easy to fragment (Shafir et al., 2001; Veron and Stafford-Smith, 2000). In addition, the *S. pistillata* is a rather resilient coral, more often than not, taking a hold in areas with harsher conditions in the reef as well as having a rather fast growth rate, thus making it a pioneer species (Loya, 1976). These facts make the *S. pistillata* a good candidate for growth and maintenance in a lab, thereby a "guinea-pig" for coral studies.

Regeneration

As in other cnidarian taxa (Fujita et al., 2021; Technau and Steele, 2011) one of the biological hallmarks of corals is their regeneration power, as coral colonies are capable to repair lesions of various sizes (Henry and Hart, 2005) to amend polyp lost (Henry and Hart, 2005; Lirman, 2000; Meesters et al., 1997a), and to regenerate the whole body (colony), even from minute fragments (Luz et al., 2018, 2021; Shafir et al., 2003, 2006). These regenerative traits allow colonies to stand a whole plethora of damages inflicted by physical and biological drivers, primarily the results of (a) fish (e.g., parrotfish and pufferfish) bites, that snip off pieces of tissues and skeletons (Cole et al., 2008), (b) predation by invertebrates, such as snails and starfish, and corallivorous fish (e.g., butterflyfish), which consume just the tissue, leaving the skeleton intact (Baums et al., 2003; Cole et al., 2008) and (c) forceful dynamics of storms, macroalgae and human activities that might cause mechanical damages such as colonial

fragmentation (Dollar and Tribble, 1993; Harmelin-Vivien, 1994; Rogers et al., 1983) or tissue abrasion (Harmelin-Vivien, 1994; Manikandan et al., 2021).

The power for regeneration would require the proper reconstruction of both, the tissues and the skeletal frameworks, in a highly demanding process, requiring the provision of cells and energy resources (Meesters et al., 1996; Oren et al., 1997; Rinkevich, 1996), as well as the proper landscape for pattern formation (Guerrini et al., 2021; Shaish and Rinkevich, 2009; Shaish et al., 2006). The regeneration would preferably be a fast process, as damaged tissue with long-term exposed skeleton may expose the colony to harsh competition and diseases through opportunistic settlement of turf organisms including invertebrates, algae, and bacteria (Diaz-Pulido et al., 2009; Mullen et al., 2004; Rodríguez et al., 2009; Voss and Richardson, 2006), thus, expressing the dire need of the coral to fast recovering of its wounded tissues and skeletal frameworks.

Wound healing and regeneration

The regenerative power of anthozoans was identified over a century ago (Child, 1903) and studied intensively on the morphological level. Yet, not much is known about the cellular patterns and the molecular machinery by which a coral colony would regenerate lost tissues and polyps and heal wounds. Most studies examined the regeneration rates, rates by which denuded surface areas were recovered and time periods for complete repair of wounds at certain sizes (Eh et al., 1997; Hall, 1997; Henry and Hart, 2005; Lirman, 2000; Meesters and Bak, 1993, 1995; Meesters et al., 1994, 1997a, 1997b; Oren et al., 1997; Titlyanov et al., 2005). An equation developed by Meesters et al. (1997a) based on observations of wound healing in corals in their natural environment and logistic growth model, could quite accurately predict the time it will take a colony to fully repair wounds in varying sizes and calculate the surface area recovered by the colony at each point during the wound healing process. Recent studies have further followed coral regeneration under the scope of various environmental stress conditions (Bonesso et al., 2017; Counsell et al., 2019; Kaufman et al., 2021) and in corroboration with the physiological statuses of the coral tissues (Horricks et al., 2019, 2020). Yet, no research has focused in elucidating the most promising building blocks for coral regeneration, the putative coral adult stem cells (ASCs) and their progenitors.

Stem cells

Within the phylum Cnidaria, the hydrozoans represent the most well-studied taxon for ASCs, mastered by the model species *Hydra*, a taxonomic class of organism possessing three lines of ASCs, two unipotent epithelial stem cells (the epidermal and endodermal stem cells) and the multipotent interstitial cells, the I-cells (Bode, 2003; Bosch, 2009; Gierer et al., 1972; Gold and Jacobs, 2013; Vogg et al., 2019). The most studied I-cells constitute a pool of migratory, undifferentiated cells, capable of self-renewal and differentiation into various specialized cell types, including the germline, while the two epithelial stem cell lineages are capable of self-renewal of only the appropriate epithelial layer (Vogg et al., 2019), altogether allowing the hydrozoans' unprecedented regenerative power. In the Scyphozoa, no I-cells exist and another cell type, the amoebocytes, is probably the source of stem/progenitor cell proliferation during regeneration, in addition to trans/dedifferentiation processes (Fujita et al., 2021; Lesh-Laurie et al., 1991; Nakanishi et al., 2008). The amoebocytes feature irregular shapes, are found in both epithelia and within the mesoglea and commute using filopodial extensions. Yet, almost nothing is known about ASCs in the Anthozoa, where some studies have claimed for the appearance of amoebocytes conjoining with regenerating wounds, although their specific roles in tissue regeneration is still under dispute (Hutton and Smith, 1996; Meszaros and Bigger, 1999; Mydlarz et al., 2008; Olano and Bigger, 2000; Palmer et al., 2011; Patterson and Landolt, 1979; Vargas-Ángel et al., 2007). Just a single study (Raz-Bahat et al., 2006) has elucidated an unorganized cell masses at the margins of a growing tissues from nubbins of the branching coral *Stylophora pistillata*. These cells, residing at the very front of tissue growth, were described as small, round cells with pronounced nuclei, the typified morphology of stem cells.

In order to identify stem cells across various taxa, many markers are used (Alié et al., 2015; Avasthi et al., 2008; Gonzalez et al., 2015; Juliano et al., 2014; Shibata et al., 1999; Shukalyuk et al., 2007). These markers may include a wide range of proteins and mRNA sequences identifying with stemness properties. In well studied organisms, such as rats, mice and human, a wide array of membrane proteins unique to different types of stem cells, typically used for stem cell identification. These may include CD24, CD34, CD41, CD44, etc… (Ianez et al., 2013; Mannelli et al., 2015; Okudela et al., 2012; Zhao et al., 2012). While constituting as very prominent and trustworthy markers in mammalian model organism, these markers are poorly studied in non-model organisms. Nevertheless, a group of well-known and rather conserved proteins, directly related to cell potency and self-renewal properties is widely used to ascertain the stemness of a cell. This group includes chiefly PIWI, VASA, PL10 and Bruno (Gonzalez et

al., 2015; Guo et al., 2006; Juliano et al., 2014; Sekiné et al., 2015; Shibata et al., 1999, 10; Shukalyuk et al., 2007). Those proteins take place in various stem cell processes and are widely conserved in the eumatozoa. They could be recognized within a cell by identifying the protein itself, using specifically designed antibodies, or by their transcription, as mRNA.

As mentioned before, PIWI is one of the most prominent subfamily of proteins used for stem cell identification in many model and non-model animals (Gonzalez et al., 2015; Juliano et al., 2014; Rinkevich et al., 2010). It is well studied in various animals, as well as plants (Vaucheret, 2008), from different phyla (Cox et al., 1998; Funayama et al., 2010; Girard et al., 2006). It is a RNA binding protein belonging to the Argonaute/Piwi family. It was initially identified in Drosophila, where it had been found to have as significant role in slicing miRNAs to use as post-transcriptional silencing or siRNAs for both transcriptional and post-transcriptional silencing (Cox et al., 1998; Gonzalez et al., 2015). In both germline and somatic stem cells PIWI is crucial for cell's self-renewal (Gonzalez et al., 2015). Usually, in order to estimate the amount of proliferating cells in a tissue, BrdU (5-bromo-2'-deoxyuridine) labelling or specific cell-cycle markers, such as Phospo-Histone H3 or PCNA are being used.

The aim of this study is to follow on the whole organismal level (including dynamics, patterns and rates) and on the cellular level the regeneration responses to injuries inflicted upon fragments of *S. pistillata.* A unique assay, utilising fragments of three genets of *S. pistillata*, was used to better understand this process of tissue regeneration and growth. Mechanical circular wounds, 1.5 mm in diameter, were inflicted on regrown 2D tissue and skeleton over a glass slides. We've hypothesised that all the three genets will show similar regeneration trends and rates. Also, we've expected that the cells appearing in the foremost part of the regenerating tissue (as indicated by Raz-Bahat et al., (2006) will express stemness markers as well as proliferation markers, thereby suggesting that they're putative ASC. Total regeneration time for each genet and success in performing full regeneration, as well as regeneration rate of each wound were calculated. In addition, cross-sections were attained along the tissue regeneration and growth process which were stained with hematoxylin & eosin, as well as specific antibodies for Piwi protein (stemness marker) and pH3 (proliferation marker). Also, gene expression of seven stemness markers: *Piwi1*, *Nanos-1* homologue, *Tudor5-like*, *Tudor7-like*, *Boule-like isoform*, *Sox2-like* and *Myc1-like* was examined in regenerating, fully healed and growing in comparison with naïve tissue.

Article draft:

Coral tissue growth and regeneration conjoin the advent of adult stem cell-like cells

Jonathan Levanoni[1,2,*], Amalia Rosner[1], Ziva Lapidot[1], Buki Rinkevich[1]

[1]Israel Oceanographic and Limnological Research, National Institute of Oceanography, Haifa 3109701, Israel

[1]Marine Biology department, Leon Charney School of Marine Sciences, Haifa University, Haifa, Israel

* Author for correspondence (j.levanoni@gmail.com)

Summary statement

This study presents for the first time in a scleractinian coral stem-cells like and stemness expression in regenerating tissues and naturally fast-growing tissues, all within unique epidermal extensions.

Abstract

Although regeneration processes in the Anthozoa have been intensively studied for more than a century, the cellular mechanisms by which tissue regeneration are executed are not understood and the putative adult stem cells (ASC) are undisclosed. Here we aimed to elucidate morphological, cellular and some gene expression patterns in injured spreading tissues of nubbins from the branching coral *Stylophora pistillata* and in the margins of fast-growing tissues. Tissue regeneration following small circular incisions (each 2.77mm^2), where a single polyp had been removed, was followed using microphotography, histological staining, pH3 and Piwi immunohistochemistry and qPCR for gene expression seven stemness markers: *Piwi1*, *Nanos-1* homologue, *Tudor5-like*, *Tudor7-like*, *Boule-like isoform*, *Sox2-like* and *Myc1-like*. Regeneration of the small injuries was completed within 9 days in 84.5% (n=64) of the assays. In nearly 35% of the cases a single polyp was regrown within this time, while in 60% no polyp regrew and in about 6% more than one polyp appeared on the regenerating margins. Results also revealed the formation of an 'addendum', an ectodermal epithelium in growing edges (in

regenerating and normal growing fronts) inhabited by a cluster of small cells featuring dense nuclei, resembling ASCs. Many of these cells express pH3 as well as Piwi proteins. Most of the stemness marker genes were upregulated in newly regenerated tissue or in the spreading coral tissue. This study revealed for the first time in the Anthozoa the presence and participation of ASC-like cells in regenerating tissues and in naturally growing tissue fronts.

Keywords: Tissue regeneration, coral, stem cell, wound healing, polyp, Stemness

Introduction

As in other cnidarian taxa (Fujita et al., 2021; Technau and Steele, 2011) one of the biological hallmarks of corals is their regeneration power, as coral colonies are capable to repair lesions of various sizes (Henry and Hart, 2005) to amend polyp lost (Henry and Hart, 2005; Lirman, 2000; Meesters et al., 1997a), and to regenerate the whole body (colony), even from minute fragments (Luz et al., 2018, 2021; Shafir et al., 2003, 2006). These regenerative traits allow colonies to stand a whole plethora of damages inflicted by physical and biological drivers, primarily the results of (a) fish (e.g., parrotfish and pufferfish) bites, that snip off pieces of tissues and skeletons (Cole et al., 2008), (b) predation by invertebrates, such as snails and starfish, and corallivorous fish (e.g., butterflyfish), which consume just the tissue, leaving the skeleton intact (Baums et al., 2003; Cole et al., 2008) and (c) forceful dynamics of storms, macroalgae and human activities that might cause mechanical damages such as colonial fragmentation (Dollar and Tribble, 1993; Harmelin-Vivien, 1994; Rogers et al., 1983) or tissue abrasion (Harmelin-Vivien, 1994; Manikandan et al., 2021)

The power for regeneration would require the proper reconstruction of both, the tissues and the skeletal frameworks, in a highly demanding process, requiring the provision of cells and energy resources (Meesters et al., 1996; Oren et al., 1997; Rinkevich, 1996), as well as the proper landscape for pattern formation (Guerrini et al., 2021; Shaish and Rinkevich, 2009; Shaish et al., 2006). The regeneration would preferably be a fast process, as damaged tissue with long-term exposed skeleton may expose the colony to harsh competition and diseases through opportunistic settlement of turf organisms including invertebrates, algae, and bacteria (Diaz-Pulido et al., 2009; Mullen et al., 2004; Rodríguez et al., 2009; Voss and Richardson, 2006), thus, expressing the dire need of the coral to fast recovering of its wounded tissues and skeletal frameworks.

The regenerative power of scleractinian corals was identified over a century ago (Child, 1903) and studied intensively on the morphological level. Yet, not much is known about the cellular patterns and the molecular machinery by which a coral colony would regenerate lost tissues and polyps and heal wounds. Most studies examined the regeneration rates, rates by which denuded surface areas were recovered and time periods for complete repair of wounds at certain sizes (Eh et al., 1997; Hall, 1997; Henry and Hart, 2005; Lirman, 2000; Meesters et al., 1994, 1997a, 1997b; Meesters and Bak, 1993, 1995; Oren et al., 1997; Titlyanov et al., 2005). Recent studies have further followed coral regeneration under the scope of various environmental stress conditions (Bonesso et al., 2017; Counsell et al., 2019; Kaufman et al., 2021) and in corroboration with the physiological statuses of the coral tissues (Horricks et al., 2019; Horricks et al., 2020). Yet, no research has focused in elucidating the most promising building blocks for coral regeneration, the putative coral adult stem cells (ASCs) and their progenitors.

Within the phylum Cnidaria, the hydrozoans represent the most well-studied taxon for ASCs, mastered by the model species *Hydra*, a taxonomic class of organism possessing three lines of ASCs, two unipotent epithelial stem cells (the epidermal and endodermal stem cells) and the multipotent interstitial cells, the I-cells (Bode, 2003; Bosch, 2009; Gierer et al., 1972; Gold and Jacobs, 2013; Vogg et al., 2019). The most studied I-cells constitute a pool of migratory, undifferentiated cells, capable of self-renewal and differentiation into various specialized cell types, including the germline, while the two epithelial stem cell lineages are capable of self-renewal of only the appropriate epithelial layer (Vogg et al., 2019), altogether allowing the hydrozoans' unprecedented regenerative power. In the Scyphozoa, no I-cells exist and another cell type, the amoebocytes, is probably the source of stem/progenitor cell proliferation during regeneration, in addition to trans/dedifferentiation processes (Fujita et al., 2021; Lesh-Laurie et al., 1991; Nakanishi et al., 2008). The amoebocytes feature irregular shapes, are found in both epithelia and within the mesoglea and commute using filopodial extensions. Yet, almost nothing is known about ASCs in the Anthozoa, where some studies have claimed for the appearance of amoebocytes conjoining with regenerating wounds, although their specific roles in tissue regeneration is still under dispute (Hutton and Smith, 1996; Meszaros and Bigger, 1999; Mydlarz et al., 2008; Olano and Bigger, 2000; Palmer et al., 2011; Patterson and Landolt, 1979; Vargas-Ángel et al., 2007). Just a single study (Raz-Bahat et al., 2006) has elucidated an unorganized cell masses at the margins of a growing tissues from nubbins of the branching coral *Stylophora pistillata*. These cells, residing at the very front of tissue growth, were described as small, round cells with pronounced nuclei, the typified morphology of stem cells.

By following the regeneration of injured tissues in *S. pistillata* at the whole organismal level (including dynamics, patterns and rates) and at the cellular level, this study aims to establish the morphological/cellular processes for regeneration. Results confirmed morphological patterns for regeneration allied with cellular events and the expression of proliferation and germ-line multipotency program (GMP) genes (Alié et al., 2015) associated with the regeneration process, thus suggesting for stem cell involvement in corals' regeneration processes.

Materials and methods

Collection and maintenance

Three colonies of *S. pistillata* that were naturally recruited from the plankton and grew on the frames of a floating coral nursery, about 300 metres off the Northern Beach in Eilat, Israel (N 29° 32' 34.843" E 34° 58' 24.88"; 9-11 m depth), were collected, marked as colonies 1 to 3 and translocated to the running water tables facility at the Israel Oceanography & Limnological Research (IOLR, Haifa). The water tables were supplied with seawater pumped from the adjacent Mediterranean shoreline (about 1 m deep), and maintained under constant 24°C temperature (using a chiller unit) and under light regime of 12:12 h (light: dark), using metal halide spotlights (400 W, 10000 K and 14000 K) (Bockel and Rinkevich, 2019).

Fragmentation and lateral growth

Following an acclimation period of at least two weeks, the coral colonies were fragmented to nubbins (ca.160 nubbins per colony), each constituting at least ten polyps, using either wire cutters or diamond plated band saw (for thicker parts of the colony), as described (Shafir et al., 2001). Then, the nubbins were attached to glass slides (two per slide), using Loctite superglue (Shafir et al., 2001, 2003, 2006). The slides were placed four in each proprietary plastic rack and into 30 litre aquaria (similar conditions to the water tables). The nubbins were left to grow laterally on the substrates, as described (Shafir et al., 2001; Shafir et al., 2003), and encrusted the slides until tissue extended at least 1cm from nubbin's base, and skeleton was formed underneath the expanded tissue (Fig. 1). The slides were cleaned with a sharp blade once a week to remove debris, algae and pests.

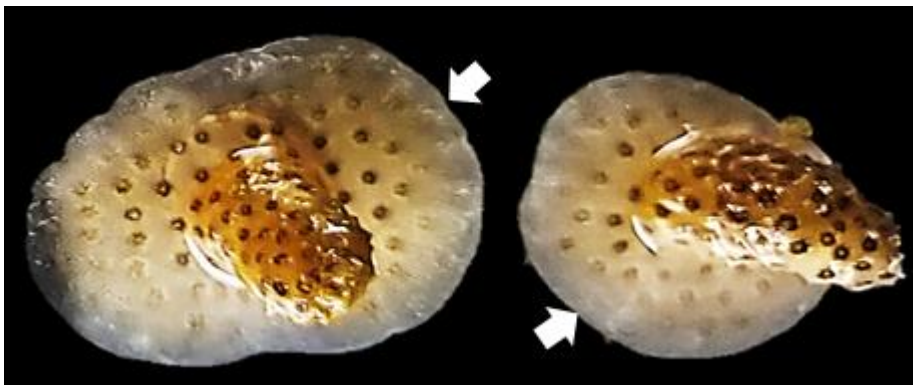


Fig 1. Two *S. pistillata* nubbins with grown and 2D spread tissue and skeleton over the glass slide ensuing about 70 days of growth following fragmentation (margins indicated by white arrows).

Regeneration assays

For the regeneration assays, 34 nubbins of each genet were allocated to separate aquaria under similar conditions (one for each genet). Regeneration assays of circular injuries were performed by punching holes through the tissues and skeletons using 1.5mm diameter dermal punch, removing completely a single polyp and its intimate surrounding coenosarc. Then, the nubbins were placed back into the aquaria and each was photographed once a day using a Nikon SMZ1000 dissection microscope coupled with DeltaPix Invenio 3SII microscope mounted camera. Calibration was performed by photographing a scale attached to a similar glass slide before every daily photographing session. Daily photographing seized when all injuries either completely healed or seized regenerating altogether. For each time point, the injury surface area was calculated using ImageJ 1.52a. The genet specific regeneration coefficient was calculated as the percentage of injuries fully healed. Newly established polyps within the regenerating area or surrounding the injuries were indicated.

Histology

Nubbins (n=11) from genet no 1 were used for the regeneration assays. One nubbin was sacrificed (by fixation) at injuring time and every 24 h thereafter, until 9 days post injury (dpi). One intact nubbin served as control. Fixation was performed using Bouin's fixative (15:5:1; picric acid, formalin and acetic acid, respectively) for 2 h. The skeletons were decalcified following Rinkevich and Loya (1979) and staining with hematoxylin and eosin followed Rinkevich et al. (2010). Five slides from each time point (n=55) were stained photographed and analysed using Nikon DS-Fi2 microscope camera coupled with Nikon Eclipse Ni-U upright light microscope.

Immunohistochemistry - Phosphorylated Histone H3, Piwi

Twelve random slides (five per antibody, pHH3 and Piwi, and one negative control for each) from each time point (n=10) were used. Immunohistochemistry protocol followed Rinkevich et al. (2010). Antibodies used: mouse anti-Phosphorylated Histone H3 monoclonals (1:50, Santa Cruz sc-374669), rabbit anti-Bl (*Botrylloides leachi*)-Piwi polyclonals (1:3000; Rinkevich et al., 2010)). Secondary antibodies used: Jackson Laboratories goat anti-mouse (1:300, 115-605-003) and goat anti-rabbit (1:300, 111-605-003), both coupled with Alexa-647. The slides were counterstained with 4',6-diamidino-2-phenylindole, dihydrochloride, DAPI (1 µg ml^{-1}). Cover slips were mounted atop the slide using Sigma-Aldrich Flouromount.

Stained sections were photographed and analysed using above mentioned microscope setting. For each slide, we followed the location and morphology of the stained cells in close vicinity to the wound and to the regenerating edge.

Quantitative PCR

S. pistillata nubbins (n=45), 15 of each of three genets, were allocated to a 30L aquarium supplied with continuous running seawater and were subjected to regeneration assays, as described. Tissues were collected using a scalpel and pooled in one tube for each genet at four time points: (1) tissues removed at onset, as part of the regeneration assay, were assigned as RS (resting state) (n=10 per genet), (2) regenerating areas at 10 hours post injury assigned as Reg (n=5 per genet), (3) additional fully healed injuries assigned as FH (n=5 per genet) and (4) tissue from the peripheries of the laterally grown tissue's edges of control, unharmed nubbins assigned as TE (n=5 per genet). Pooled tissues were fixed in RLT buffer (supplied with QIAGEN RNeasy kit) and stored at -70°C. Total mRNA was extracted using RNeasy Mini Kit (Cat. No. 74104) and RNase-free DNase kit Cat. No. 79254; Qiagen, Germany). cDNA was synthesised using RevertAid, First Strand cDNA Synthesis Kit (cat no. K1622; Thermo Scientific, https://www.thermofisher.com/il/en/home.html) according to manufacturer instructions.

The mRNA expressions of seven stemness genes were studied (*Piwi1, Nanos-1 homologue, Tudor5-like, Tudor7-like, Boule-like isoform, Sox2-like and Myc1-like*; https://www.ncbi.nlm.nih.gov/genbank/; accession numbers in Table S1) in the four biological states using the relative qPCR method. Two normalising genes were utilised (*EIF4A1* and *18S*; Suppl. Table 1). qPCR analyses were performed with Fast SYBR™ Green Master Mix (Thermo Fisher Scientific, MA, USA) according to manufacturer instructions using a StepOnePlus Real-Time PCR System (Thermo Fisher Scientific, MA, USA). The $\Delta\Delta C_T$ method was used to analyse the relative (fold) changes of the tested genes' expressions in the treated tissues versus untreated controls, as described (Pfaffl, 2001). Statistical analyses were carried out using IBM SPSS 25.0 on three biological replicates. Differences were calculated between sets of ΔC_T (threshold Cycle) of examined, following normalisation using the normalising genes. Then, analyses were carried on by performing repeated-measures ANOVA and LSD (Fisher's Least Significant Difference) and post-hoc to compare between the four biological states. In cases in which data's sphericity couldn't be assumed using Mauchly's Test of Sphericity, Huynh-Feldt Correction was used to determine the significance of the test.

Results

Morphological outcomes

Wound healing and regeneration processes were followed up to 9 dpi, when most injuries were fully regenerated (Fig. 2A-J; Table 1). Within this time period, tissue regrew into the wounded tissue deprived site, where an active polyp resided. In most cases the tissue grew in nearly symmetrical manner from the circular wounded edges towards the wound centre. Out of 75 assays studied for a period of 9 days, in 11 (15.5%; Table 1) wounds were not fully regenerated. The major cause for incomplete regeneration during this period was algal growth within the wound site (Fig. 2K-T). The percentages of fully regenerating nubbins per colony varied between 73.3% to 96.3% (Table 1) and was in average 84.5% (Table 1).

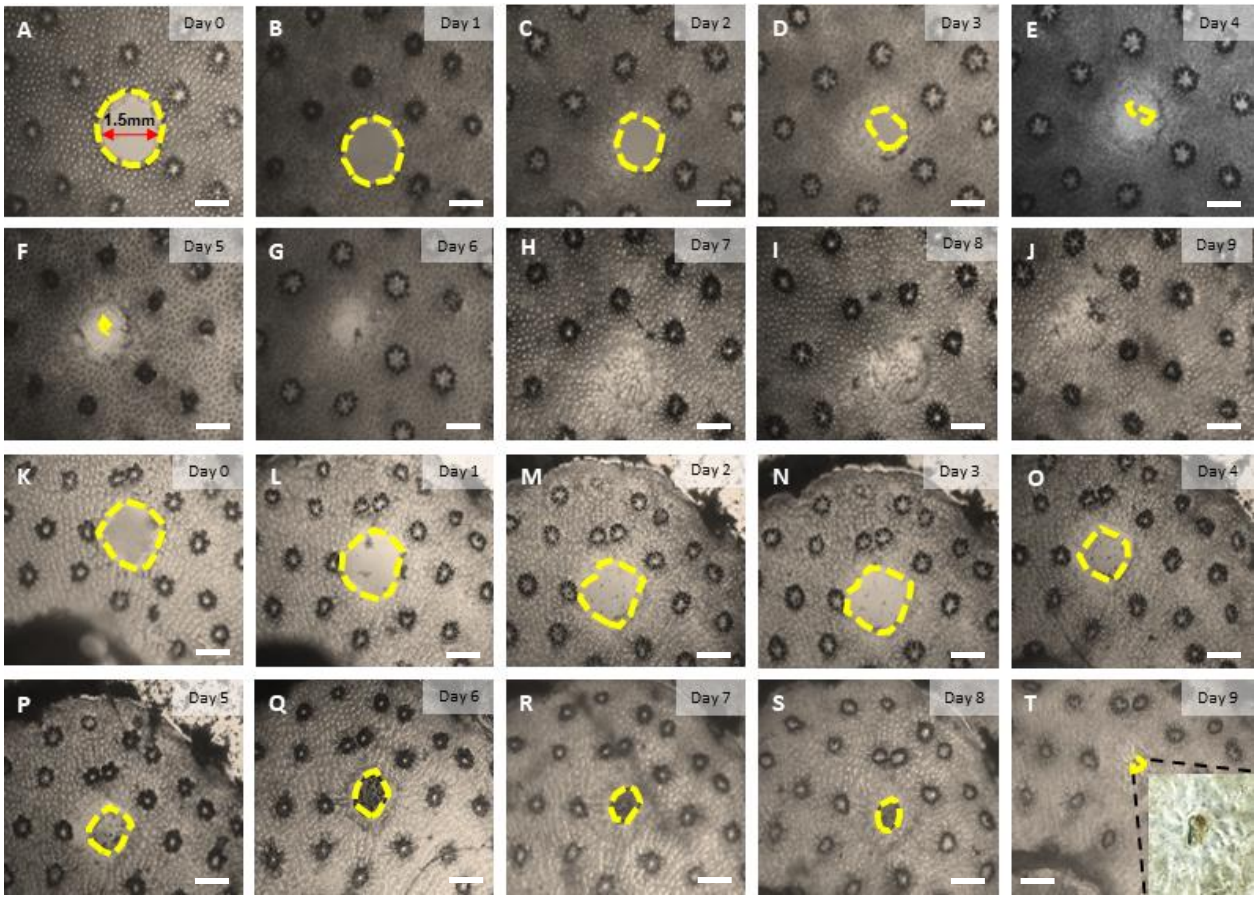


Fig. 2. *Stylophora pistillata* regenerating nubbins. Daily images of wound areas in two regeneration assays, followed up to 9 days from onset. Both wounds were circular, each 1.5mm diameter. A-J: Fully regenerated case by day 6; K-T: Partial regeneration. Algal growth in the wound area. Yellow dashed lines indicate the wound's edge. Note the 'pale' contour of the regenerated tissue, indicating the tissue front. Scale bars = 1mm

Table 1. The distribution of fully regenerating assays (at 9 days post injury) in the three studied *S. pistillata* genet.

Genet	Assays (n)	Regeneration not completed (n)	Regeneration completed	
			(n)	(%)
1	18	2	16	88.89
2	27	1	26	96.29
3	30	8	22	73.33
Total	75	11	64	84.51

A complete (100%) coverage of the regenerated areas was achieved by 9 dpi (Fig. 3A), irrespective of the coral genet. Differences in genets' regeneration rates were noted, yet the three curves share a similar non-linear logistic growth trajectory. Total regeneration rates (start to finish) were calculated for each genet separately (Fig. 3B), revealing that regeneration rates of genet 2 are significantly higher than genet 1 and 3, which are similar (Kruskal-Wallis H test: X^2=13.359, p=0.001).

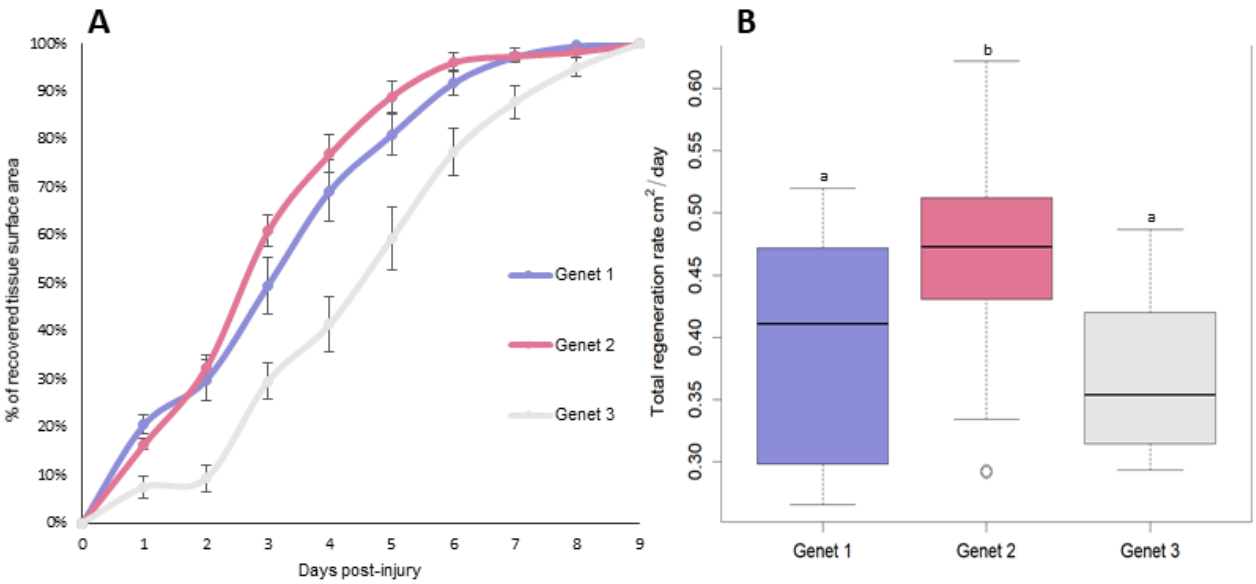


Fig. 3. Differences between the three colonies (genet 1-3) along the regeneration process. (A) Percentage (±s.e) of regenerated surface area along nine days from injury. (B) Calculated regeneration rates for all fully regenerated injuries in each genet. Similar letters indicate non-significant differences (Mann-Whitney U-test, p<0.05)

In the regeneration assay a whole targeted single polyp was removed. Yet, at day 9, in nearly 60% (n=38) of the cases where full regeneration was accomplished (n=64), no new polyp re-appeared (Fig. 4A, Table 2), while in the other 26 cases 1-3 silhouettes of new polyps emerged in close vicinity to the closed wound (Fig. 4B-D, Table 2), potentially defying the *S. pistillata* typical polyps' pattern formation (Guerrini et al., 2021). Statistical significance (χ^2=27.125, p<0.001, d.f.=2) is emerged when comparing the number cases in which 0 polyps regrown, 1 new polyp regrew and 2 and 3 combined.

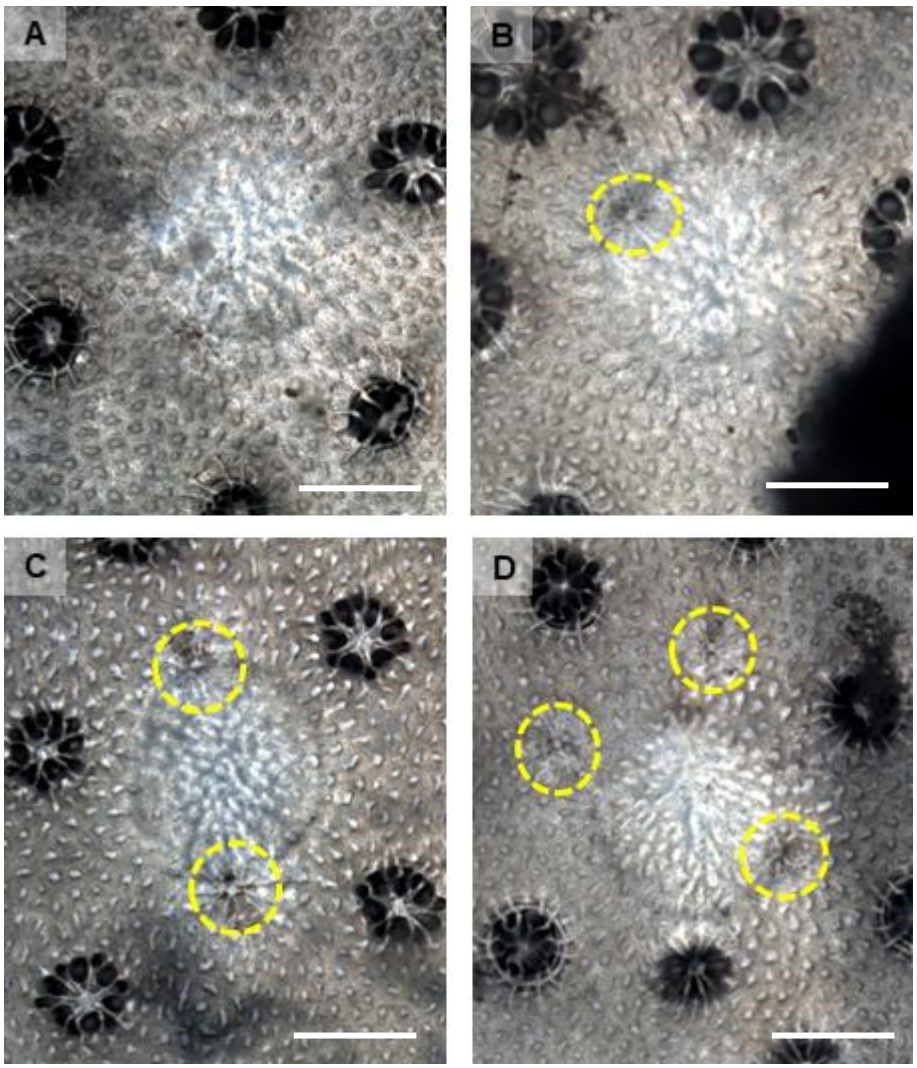


Fig. 4. Fully regenerated tissue of *S. pistillata* at 9 dpi. (A) no polyp in the regenerating area; (B-D) 1-3 silhouettes of new polyps appeared (yellow dashed circles). Scale bars = 1mm

Table 2. Distribution of number of polyps emerged by 9dpi in fully regenerated assays for each genet.

# of polyps \ genet	1	2	3	Total
0	13	9	16	38
1	3	13	6	22
2	0	3	0	3
3	0	1	0	1
Total	16	26	22	64

Histological observations

Histological sections performed on tissues at onset (Fig. 5A) revealed the typical two layered continuous epithelia, including the polyps and the coenosarc, with more zooxanthella inhabiting the downward facing gastrodermis layer than the upward gastrodermis (Bockel and Rinkevich,

2019) and between-polyps connecting coenosarc cavities of gastric extensions. Due to the histological methodology, the calicoblastic epithelium is barely noticeable, and when appeared, noted scattered cells at the lowest part of the coral tissue. Upon inflicting the injury, the tissue's continuity is cut sharply (Fig. 5A, an arrow), disconnecting opposing tissue sides and the two endodermal layers that are exposed to the environment (depicted in the illustration in Fig. 5I, d0).

In the first couple of days following the injury (1-2dpi, Fig. 5B, C), 'sealing' of the distal wound's interfaces along the injury plan is documented (presumably resulted from peripheral contractions of exposed epithelia), revealing the restructuring of the epidermal epithelium that folds over the wound's edges, connecting the upper epidermal epithelium and the underneath, distal calicoblastic layer (connection point indicated with an asterisk, Fig. 5C). While tissues at proximal positions to the wound site, feature columnar ectodermal cells overlying the endodermal epithelium layer, in most distal wound sites, the epidermal epithelium is replaced by an irregular cell mass that does not match the typical single-layer, columnar cells of the epidermal epithelium (schematically illustrated in Fig. 5I parts d1-d2).

Following the initial steps of tissue regeneration (3dpi onwards, until full tissue regeneration is achieved), the histological sections documented the emergence of a novel morphological structure at the foremost extending parts of the regenerating ectodermal epithelium, morphologically depicted by condense masses of small nuclei (Fig.5C, G indicated as 'cm'). These epidermal expansions (termed as 'addendums'; schematically illustrated in Fig. 5I, d3-full regeneration) may extend distally up to about 150µm, and were found to possess dense aggregations of seemly undifferentiated cells, circular in shape and about 5µm across, resembling primary observations by Raz-Bahat et al. (2006). The calicoblastic layer starts proximally to the said epidermal epithelium expansion (Fig. 5C, G, indicated with asterisks). The newly formed gastrodermal epithelium has been further developed with traversing canals of the gastrovascular system, resembling the regular epithelial morphologies. At the very front of the gastrodermis closest to the epidermal epithelium expansion, the gastrodermal epithelial cells are larger, nearing 10µm across, and rounder, as opposed to the flatter typical gastrodermis epithelial cells in proximal possessions. Few zooxanthellae are found at this state, and in most histological sections they appear only 150-200µm proximally to the most distal part of the gastrodermis. In fully regenerated states (Fig. 5D), the epidermal epithelium 'addendum' recedes, the epidermal cells reveal the typical epithelium structures, and the gastrodermis is fully formed, containing zooxanthellae mostly within its downward facing side.

Spreading sheets of coral tissue/skeleton on the substrate- The nubbin's spreading front edges are composed of 50µm long addendums (Fig. 5E, H), each revealing a double-layers of an epidermal epithelium expansion, containing condense masses of nuclei. The calicoblastic layer begins stretching inwards from the proximal end of the addendum, right below the foremost part of the mesoglea and extends on the dorsal part of the colony. A thick mesoglea layer of up to about 30 µm (in comparison to thin 1-5 µm in most of the animal) is separating between the gastrodermis and epidermal 'addendum' at the foremost area of the tissue. The gastrodermal epithelial cells are thicker, and cube-like (nearing 10µm across). The gastrodermal epithelial cells become thinner and longer in more proximal positions. Zooxanthellae can be seen in the gastrodermis, ca. 20µm close to its edge.

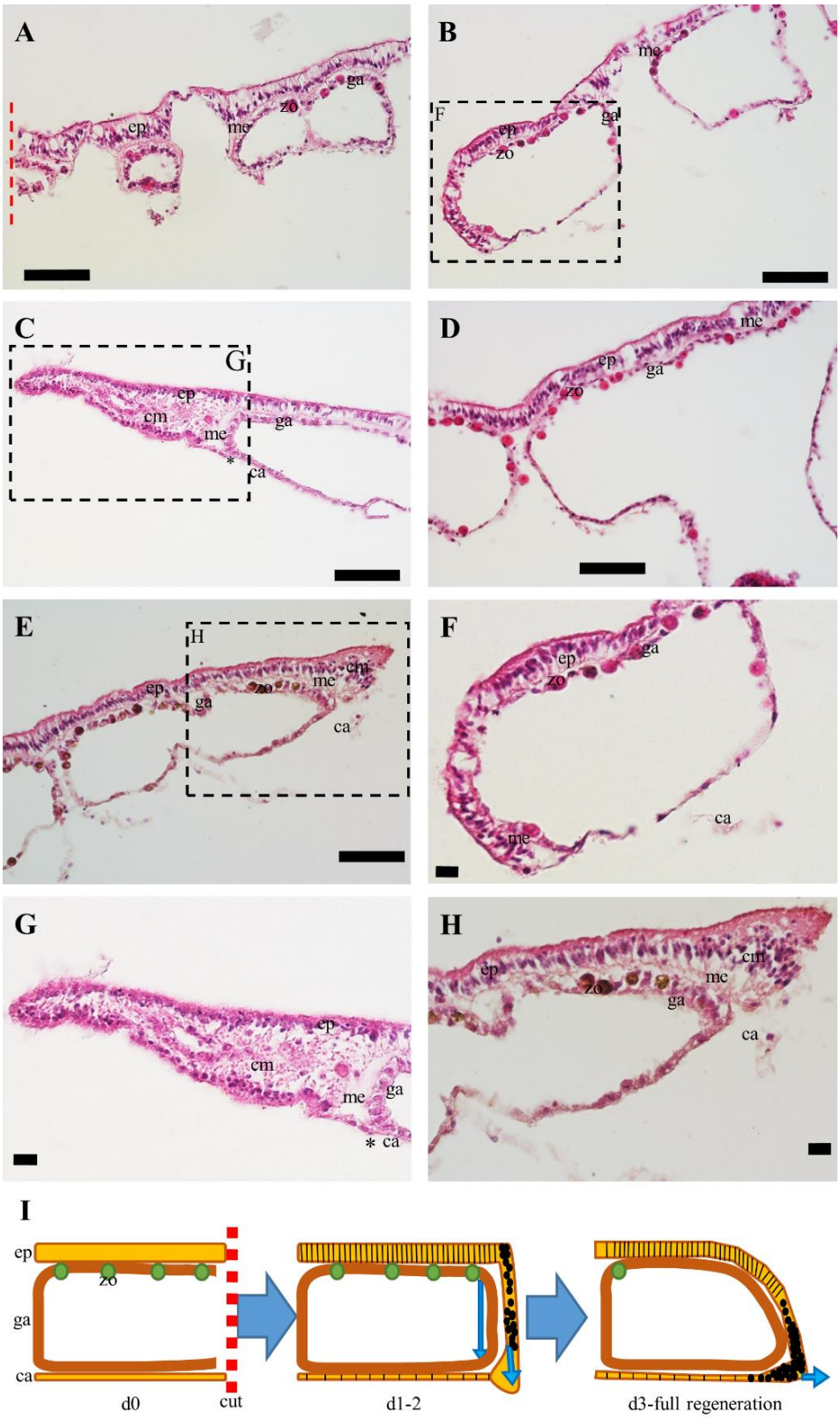


Fig. 5. *S. pistillata* cross sections at the injury sites or at spreading tissue edges, stained with hematoxylin & eosin. (A) newly injured tissue (0dpi); (B, F) tissue regrowing into the injury site at 2dpi; (C, G) tissue regrowing into the injury site at 5dpi, featuring epithelial epidermal epithelium expansion; (D) fully regenerated tissue at 6dpi, (E, H) a spreading nubbin tissue edge; (I) a schematic illustration of proposed tissue regeneration and growth process (from left to right): 0dpi- incision cutting through all tissues, 1-2dpi- initial closure of the injury and regeneration and 3dpi-full regeneration- addendum formation and fast

regeneration. (A-E) magnification X400, scale bars = 50µm, (F-H) magnification X1000, bars=10µm. ep-epidermis, ga-gastrodermis, me-mesoglea, zo-zooxanthellae, cm-cell mass, ca-calicoblastic layer, dashed red line indicates the cutting line in 0dpi (A) and asterisks indicate the most distal part of the calicoblastic layer.

Immunohistochemistry

Phospho-Histone H3 labelling

Immediately post injuring, haphazardly scattered $pH3^+$ cells were approximately equally distributed within the gastrodermal and epidermal epithelial layers (Fig. 6A), and no distinct clusters or $pH3^+$ cell aggregations were found. While at onset the cut area was almost free of $pH3^+$ cells, along the whole regeneration process many $pH3^+$ cells were seen at the injury's margins, concentrating at the very edge zones (about 100-150µm) of the epidermal epithelium layers (Fig. 6B, C), compared to only few $pH3^+$ cells appearing in the foremost parts of the gastrodermal epithelium. Following the nubbin's naturally expanding tissue on the substrate, a mass of $pH3^+$ cells is often seen in the most distal area within the epidermal epithelium (Fig. 6D). The histological observations further documented a few dozens of distinctly $pH3^+$ stained cells per section. Additionally, scattered $pH3^+$ cells were seen in the gastrodermis epithelium in more proximal positions lining the gastrovascular system (Fig. 6D), occasionally aggregating.

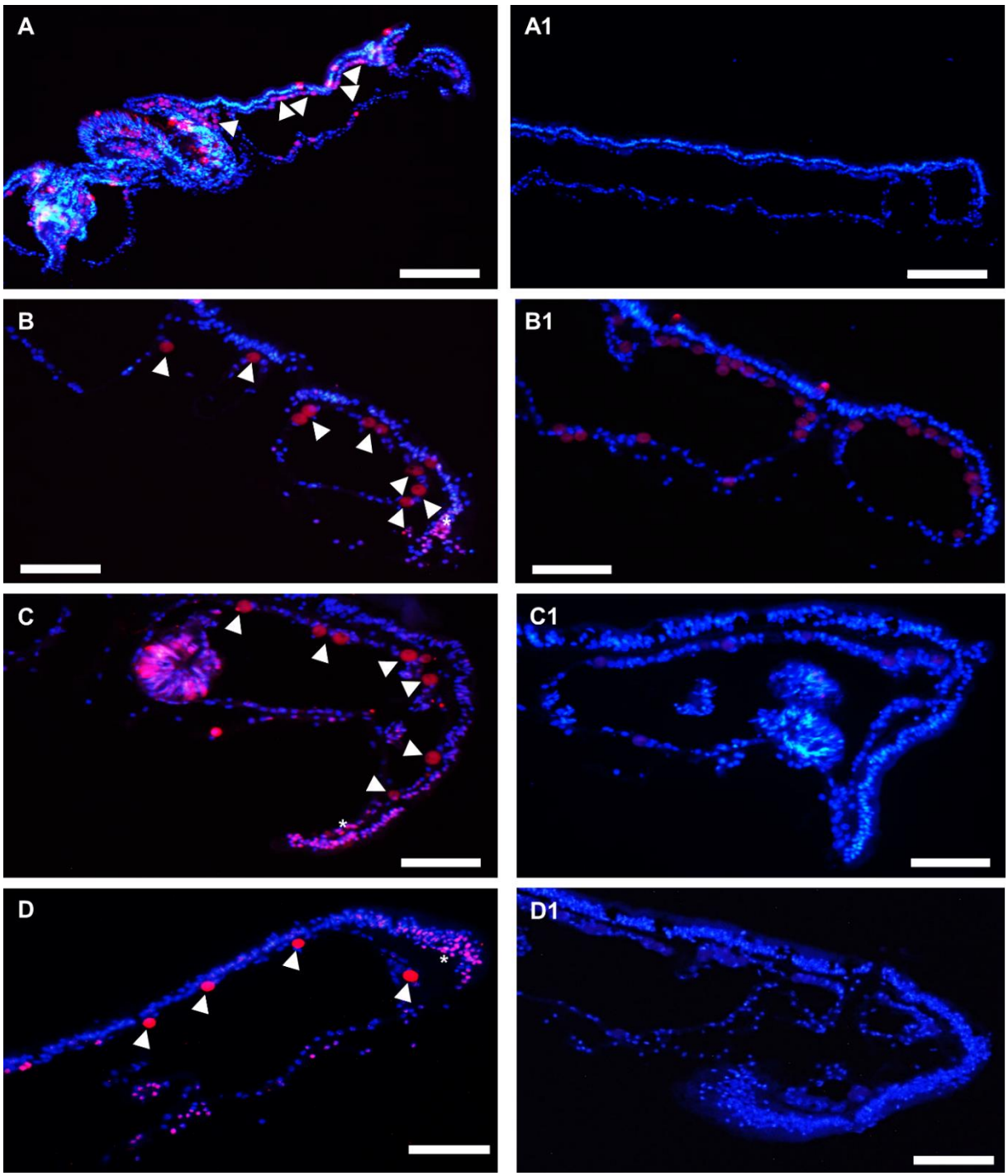


Fig. 6. *S. pistillata* cross sections at the injury sites and the naturally spreading tissue edges on the substrate, stained with anti-phosphohistone H3 (pH3$^+$) antibodies (red) and DAPI (blue). A1 to D1 are negative controls. (A, A1) newly injured tissue (0dpi); (B, B1) regenerating tissue at 2dpi) (C, C1) regenerating tissue at 5dpi, featuring epidermal epithelium expansion; (D, D1) a spreading nubbin tissue edge. (A, A1) scale bars = 500µm. (B-D, B1-D1) scale bars=50µm. Arrowheads depict the zooxanthellae. Asterisks indicate aggregates of pH3$^+$ cells.

PIWI labelling

At onset, only scattered Piwi$^+$ cells were seen in both tissue layers (also in the polyp body, and primarily in the tentacles), and no Piwi$^+$ cell clusters or cell aggregations were observed. The

Piwi protein was distinctly stained in the cells' cytoplasm, rather than in the nuclei, as recorded in cnidarians for somatic stem cells (Juliano et al., 2014). During the regeneration process and tissue rebuilding (Fig. 7B, C), aggregations of Piwi^{+} cells can be seen along the epidermal addendum, correlating with the mass of small nuclei that were observed by the H&E staining (Fig. 5B, C, F, G) and pH3^{+} cells (Fig. 6B, C). Staining was observed exclusively in the epidermal epithelia. Most of the stained cells are located in the lower epidermis, in close vicinity to the newly forming calicoblastic layer. In the spreading nubbin's tissue edge, a few Piwi+ cells were located at the very edge of the epidermal addendum of the growing front. An additional mass of Piwi+ cells can be seen proximally to them, about a 100µm apart, at the lower epidermis. Also seen in Fig. 7D are the zooxanthellae (indicated with arrowheads), autoflorescent under the employed microscope wavelength.

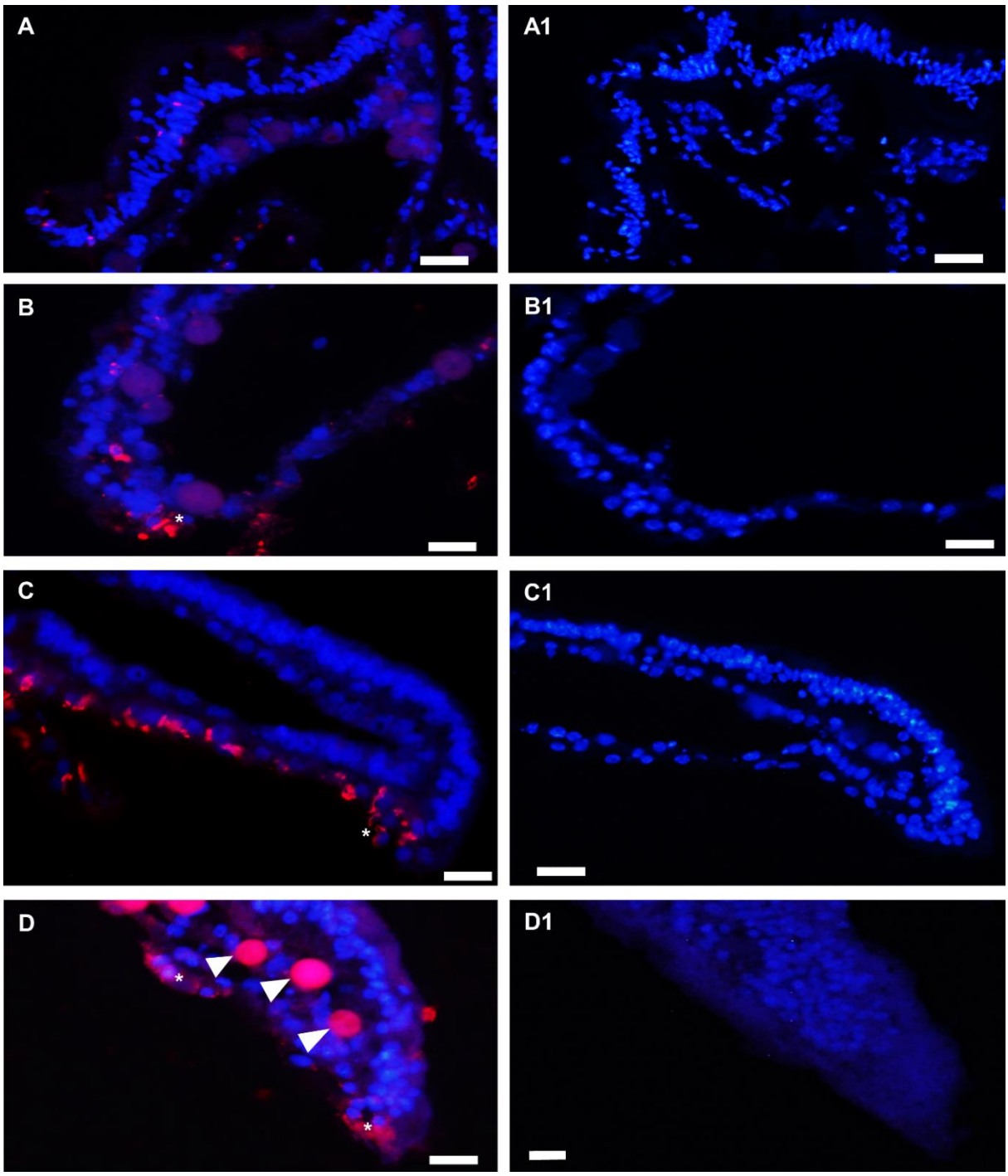


Fig. 7. *S. pistillata* cross sections at the injury sites or spreading tissue edges stained with anti-Piwi antibodies (red) and DAPI (blue). (A, A1) newly injured tissue (0dpi); (B, B1) tissue regrowing into the injury site at 2dpi;, (C, C1) tissue regrowing into the injury site at 5dpi, featuring epidermal epithelium expansion; (D, D1) a spreading nubbin tissue edge. (A1-D1) Negative controls. Magnification X1000; scale bars = 20µm; arrowheads indicate zooxanthellae and asterisks indicate Piwi^{+} cells aggregations.

qPCR analyses

mRNA expressions of the seven tested genes *(Nanos, piwi-1, boule, myc* and *Sox* orthologues and two *tudor* orthologues) showed statistically significant changes between the biological states (Fig. 8). The most prominent expression changes were recorded for the *piwi-1* gene

orthologue, whose expressions were upregulated between 123 folds at the early regeneration state (10h following injury; Reg state) to 798-folds at the fully regenerated (FR) state. In the spreading tissue edge (TE) the expression of *piwi-1* orthologue was 631 folds higher than in naïve tissues. For the *boule* orthologue the expression of the gene in all regenerative and growth states (e.g. 10hpi, fully regenerated and tissue growth edge) was upregulated significantly (8.16, 9.76 and 6.3 -fold increase, respectively) compared to the naïve tissue. In contrast, at the regeneration state (Reg) *nanos* and *myc* orthologues expressions were downregulated as compared to the RS state (3.7 and 6 -fold decrease, respectively), followed by expression increase in FR tissue in *nanos* (to 12.56 -fold increase) and TE in *myc* (1.97 -fold increase). Generally, gene expressions of six of the examined genes increased significantly either in the FR state or the TE state (except for *boule*, as described before).

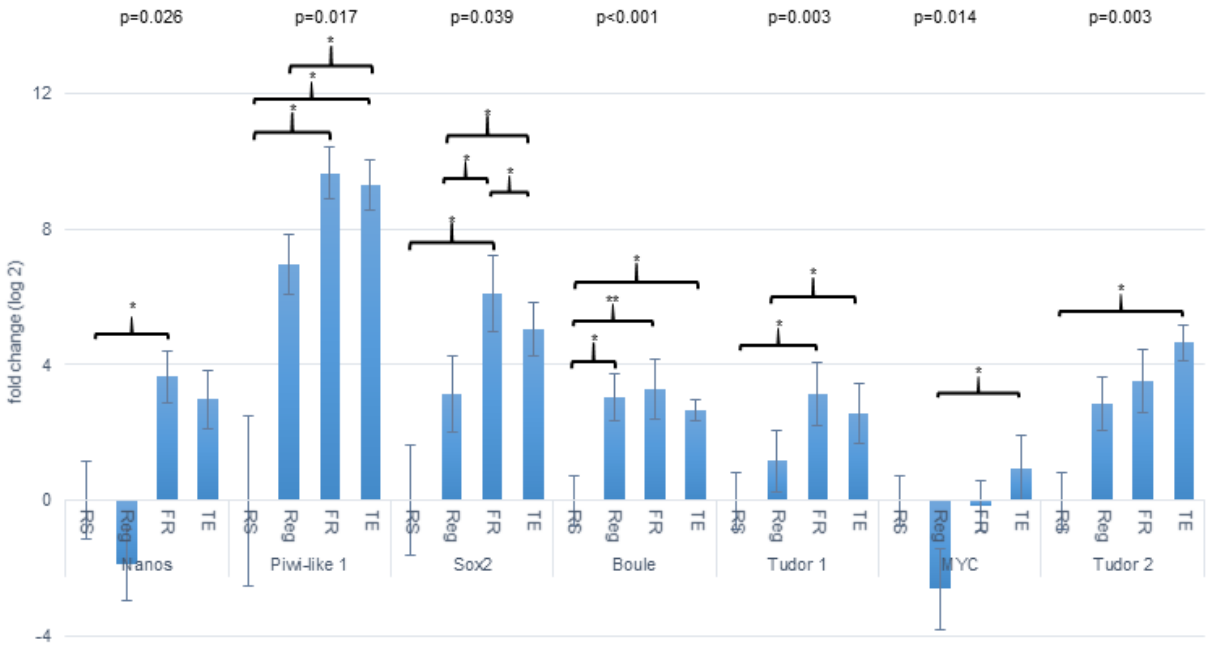


Fig. 8. Relative qPCR analyses for 7 stemness genes and transcription factors. Tissues from four biological states were sampled: naïve tissue (RS- Resting State), at onset of regeneration, Reg (10 hours post injury), FR (fully regenerated) and tissues collected from the naturally growing tissue edges (TE) of nubbins. The results are represented as fold change at log_2 scale. Repeated-measures ANOVA (p value indicated above each examined gene) and LSD (Fisher's Least Significant Difference) post-hoc *p<0.05 **p<0.001 for each state at each gene.

Discussion

This study followed tissue regeneration in small circular lesions inflicted on nubbins of the coral *S. pistillata* on the whole morphological, the cellular and gene expression levels. Within a very short period of 9 days, 84.5% of the nubbins completed regeneration, with rates characteristic to each genet studied, typified by a species-specific quadratic regression curve. While at the end the full recovery of all epithelial layers has been achieved, along with the regeneration processes, the regeneration front has been characterized by the formation of a new and distinct 'addendum' of epidermal epithelium expansion of about 150µm long, containing masses of seemingly undifferentiated cells, featuring small nuclei and expressing the well-established proliferation marker pH3, and the stemness marker Piwi, all marked putative stem cells. Along the process of tissue regeneration, we recorded changes in expression of seven stemness genes and transcription factors. Above features were shared by the most peripheral edges of the naturally spreading nubbin tissues on substrates.

Only a single polyp together with its surrounding coenosarc was removed from each participating colonial ramet, inflicting a minimal stress to the organism. Following the astogenic rules for polyps orientation in this species (Guerrini et al., 2021), it was expected to see a single polyp regenerating in the centre of the wound site for the reconstruction of polyps' pattern formation landscape, yet, unexpected results were documented. While in most (~60%) of the cases, no new polyp reappeared, all other assays resulted with the development of 1-3 polyps in the periphery of the wound, impending the typical *S. pistillata* pattern formation rules that are based on a minimal distance between adjacent polyps (Guerrini et al., 2021), and may further affect future astogenic steps. Following the rules for polyps' pattern formation as described by Guerrini et al. (2021), and the rules for new branch formation as described by Shaish et al. (2006, 2009), in cases where the tissue is completely regenerated but polyps' orientation is distorted, the colony should take a different astogenic route from the initial scheme, revealing the likelihood for exploiting these regenerating sites as centres for budding new branches. If this line of thoughts is approved in further experimentation, it may impose a novel ecological trait to coral tissue regeneration. Under this scenario, small tissue damages inflicted by corallivorous organisms and/or mechanical drivers (Baums et al., 2003; Cole et al., 2008; Dollar and Tribble, 1993; Harmelin-Vivien, 1994; Rogers et al., 1983), may affect future colony's canopy structures by shaping the coral colony branching patterns.

This study further points, for the first time in the Anthozoa, to the presumed *de novo* appearance of adult stem cells (ASCs) associated with coral regeneration and fast growth processes. ASCs

were shown to take place in myriad of biological processes in many multicellular organisms (Cervello and Simon, 2009; Fierro-Constaín et al., 2017; Juliano et al., 2014; Rink, 2013; Rinkevich et al., 2007; Shibata et al., 1999; Shukalyuk et al., 2007), particularly in tissue regeneration and growth processes. In other cnidarians, such as the well-studied *Hydra* (Hydrozoa), three types of ASCs have been identified, two epithelial, the ectodermal and endodermal unipotent stem cells and the multipotent interstitial stem cells, the i-cells (Bosch, 2009). Although being a basal group (Collins, 2009), very little is known for the existence of ASCs in the Scleractinia (Gold and Jacobs, 2013). No I-cells are recorded in the Anthozoa and the presumable stem cells, the amoebocytes, indicated in some studies (Luz et al., 2021; Meszaros and Bigger, 1999; Olano and Bigger, 2000; Palmer et al., 2011; Reyes-Bermudez and Miller, 2009) revealed just anecdotical associations with regeneration processes. Yet, cell proliferation had been found to shape and dictate anthozoans regeneration. Bisection of the sea anemones *Nematostella vectensis* has initiated high rates of cell proliferation, with no sign for any morphallactic mode of regeneration, typified to *Hydra* regeneration (Passamaneck and Martindale, 2012) and injuring the sea anemone *Exaiptasia pallida* had resulted with fast regeneration associated with early upregulation of genes involved in cellular movement and cell communication (van der Burg et al., 2020)

The present study has elucidated in regenerating parts and in the fast-growing edges of coral tissues the specific localities of (a) cells expressing stemness genes (*Nanos, Piwi-like 1, Sox2, boule, Tudor* and *Myc*) and (b) the formation of the 'addendum' ectodermal epithelium at growing edges (regenerating and normal growing fronts alike), all possessing clusters of stem cells like progenitors at undifferentiated states that express stemness (Piwi) and proliferation (pH3) properties. Although, not all of the examined genes were upregulated in the injuries' margins at 10hpi, in most cases they were upregulated when injury was fully closed. This result may stem from the fact that the complete stem cell activation phase has not been acquired at this earlier stage. Further, the upregulation of most stemness markers in newly regenerating sites may elucidate for late peak presence of putative ASCs during tissue reorganization and polyp reconstruction. Recent study (Reyes-Bermudez et al., 2021) suggested the presence of ASC in fast growing branch tips of *Acropora digitfera* based on cells' expression pattern in primary cell cultures that overexpress various stemness markers from the Nanos family of genes. Yet, a recent cell types atlas of *S. pistillata* larval/primary polyps cells (Levy et al., 2021) based on mRNA expression patterns, did not elucidated the animal's progenitors or stem cells, albeit the substantial growth and developmental processes characteristics to these early-life stages.

This study revealed, for the first time in the Anthozoa, the deduced presentation and participation of ASC-like in tissue regeneration and natural fast growth. These cells appear to congregate within epidermal epithelium extensions, termed by us as ‘addendums’. These results may shed light over the mechanisms by which scleractinian corals perform tissue/whole body regeneration and growth and will bridge the gap knowledge to create a uniform concept for tissue renewal across the cnidarian phylum.

Acknowledgements

We thank Elad N. Rachmilovitz and Omri (Shabby) Lebber-Shabbat for their assistance in collecting the coral colonies and constant support along the research, Guy Paz for his input and aid during the work in the lab, Shai Shafir for his smart advices and all the other members of the IOLR marine invertebrate lab.

competing interests

No competing interests declared.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Bibliography

Alié, A., Hayashi, T., Sugimura, I., Manuel, M., Sugano, W., Mano, A., Satoh, N., Agata, K. and Funayama, N. (2015). The ancestral gene repertoire of animal stem cells. *Proc Natl Acad Sci USA* **112**, E7093–E7100.

Baums, I. B., Miller, M. W. and Szmant, A. M. (2003). Ecology of a corallivorous gastropod, Coralliophila abbreviata, on two scleractinian hosts. I: Population structure of snails and corals. *Marine Biology* **142**, 1083–1091.

Bockel, T. and Rinkevich, B. (2019). Rapid Recruitment of Symbiotic Algae into Developing Scleractinian Coral Tissues. *Journal of Marine Science and Engineering* **7**, 306.

Bode, H. R. (2003). Head regeneration in Hydra. *Developmental dynamics: an official publication of the American Association of Anatomists* **226**, 225–236.

Bonesso, J. L., Leggat, W. and Ainsworth, T. D. (2017). Exposure to elevated sea-surface temperatures below the bleaching threshold impairs coral recovery and regeneration following injury. *PeerJ* **5**, e3719.

Bosch, T. C. G. (2009). Hydra and the evolution of stem cells. *BioEssays* **31**, 478–486.

Cervello, I. and Simon, C. (2009). Somatic stem cells in the endometrium. *Reproductive Sciences* **16**, 200–205.

Child, C. M. (1903). FORM REGULATION IN CERIANTHUS: I. THE TYPICAL COURSE OF REGENERATION. *The Biological Bulletin* **5**, 239–260.

Cole, A. J., Pratchett, M. S. and Jones, G. P. (2008). Diversity and functional importance of coral-feeding fishes on tropical coral reefs. *Fish and Fisheries* **9**, 286–307.

Collins, A. (2009). Recent Insights into Cnidarian Phylogeny. *Smithson. Contrib. Mar. Sci.* **38**,.

Counsell, C. W., Johnston, E. C. and Sale, T. L. (2019). Colony size and depth affect wound repair in a branching coral. *Marine Biology* **166**, 1–12.

Diaz-Pulido, G., McCook, L. J., Dove, S., Berkelmans, R., Roff, G., Kline, D. I., Weeks, S., Evans, R. D., Williamson, D. H. and Hoegh-Guldberg, O. (2009). Doom and boom on a resilient reef: climate change, algal overgrowth and coral recovery. *PloS one* **4**,.

Dollar, S. J. and Tribble, G. W. (1993). Recurrent storm disturbance and recovery: a long-term study of coral communities in Hawaii. *Coral Reefs* **12**, 223–233.

Eh, M., W, P. and Rpm, B. (1997). Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. *Marine Ecology Progress Series* **146**, 91–99.

Fierro-Constaín, L., Schenkelaars, Q., Gazave, E., Haguenauer, A., Rocher, C., Ereskovsky, A., Borchiellini, C. and Renard, E. (2017). The conservation of the germline multipotency program, from sponges to vertebrates: a stepping stone to understanding the somatic and germline origins. *Genome biology and evolution* **9**, 474–488.

Fujita, S., Kuranaga, E. and Nakajima, Y. (2021). Regeneration Potential of Jellyfish: Cellular Mechanisms and Molecular Insights. *Genes* **12**, 758.

Gierer, A., Berking, S., Bode, H., David, C. N., Flick, K., Hansmann, G., Schaller, H. and Trenkner, E. (1972). Regeneration of hydra from reaggregated cells. *Nature New Biology* **239**, 98–101.

Gold, D. A. and Jacobs, D. K. (2013). Stem cell dynamics in Cnidaria: are there unifying principles? *Development genes and evolution* **223**, 53–66.

Guerrini, G., Shefy, D., Shashar, N., Shafir, S. and Rinkevich, B. (2021). Morphometric and allometric rules of polyp's landscape in regular and chimeric coral colonies of the branching species Stylophora pistillata. *Developmental Dynamics* **250**, 652–668.

Hall, V. R. (1997). Interspecific differences in the regeneration of artificial injuries on scleractinian corals. *Journal of Experimental Marine Biology and Ecology* **212**, 9–23.

Harmelin-Vivien, M. L. (1994). The effects of storms and cyclones on coral reefs: a review. *Journal of Coastal Research* 211–231.

Henry, L.-A. and Hart, M. (2005). Regeneration from injury and resource allocation in sponges and corals–a review. *International Review of Hydrobiology: A Journal Covering all Aspects of Limnology and Marine Biology* **90**, 125–158.

Horricks, R. A., Herbinger, C. M., Lillie, B. N., Taylor, P. and Lumsden, J. S. (2019). Differential protein abundance during the first month of regeneration of the Caribbean star coral Montastraea cavernosa. *Coral Reefs* **38**, 45–61.

Horricks, R. A., Herbinger, C. M., Vickaryous, M. K., Taylor, P. and Lumsden, J. S. (2020). Differential protein abundance associated with delayed regeneration of the scleractinian coral Montastraea cavernosa. *Coral Reefs* **39**, 1175–1186.

Hutton, D. M. and Smith, V. J. (1996). Antibacterial properties of isolated amoebocytes from the sea anemone Actinia equina. *The Biological Bulletin* **191**, 441–451.

Juliano, C. E., Reich, A., Liu, N., Götzfried, J., Zhong, M., Uman, S., Reenan, R. A., Wessel, G. M., Steele, R. E. and Lin, H. (2014). PIWI proteins and PIWI-interacting RNAs function in Hydra somatic stem cells. *Proceedings of the National Academy of Sciences* **111**, 337–342.

Kaufman, M. L., Watkins, E., van Hooidonk, R., Baker, A. C. and Lirman, D. (2021). Thermal history influences lesion recovery of the threatened Caribbean staghorn coral Acropora cervicornis under heat stress. *Coral Reefs* **40**, 289–293.

Lesh-Laurie, G. E., Hujer, A. and Suchy, P. (1991). Polyp regeneration from isolated tentacles of Aurelia scyphistomae: a role for gating mechanisms and cell division. In *Coelenterate Biology: Recent Research on Cnidaria and Ctenophora*, pp. 91–97. Springer.

Levy, S., Elek, A., Grau-Bové, X., Menéndez-Bravo, S., Iglesias, M., Tanay, A., Mass, T. and Sebé-Pedrós, A. (2021). A stony coral cell atlas illuminates the molecular and cellular basis of coral symbiosis, calcification, and immunity. *Cell* **184**, 2973-2987.e18.

Lirman, D. (2000). Lesion regeneration in the branching coral Acropora palmata: effects of colonization, colony size, lesion size, and lesion shape. *Marine Ecology Progress Series* **197**, 209–215.

Luz, B. L. P., Capel, K. C. C., Zilberberg, C., Flores, A. A. V., Migotto, A. E. and Kitahara, M. V. (2018). A polyp from nothing: The extreme regeneration capacity of the Atlantic invasive sun corals Tubastraea coccinea and T. tagusensis (Anthozoa, Scleractinia). *Journal of experimental marine biology and ecology* **503**, 60–65.

Luz, B. L. P., Miller, D. J. and Kitahara, M. V. (2021). High regenerative capacity is a general feature within colonial dendrophylliid corals (Anthozoa, Scleractinia). *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* **336**, 281–292.

Manikandan, B., Padelkar, A. A., Ravindran, J. and Joseph, S. (2021). Histopathological investigation of the reef coral Goniastrea sp. affected by macroalgal abrasion. *Marine Biology* **168**, 1–7.

Meesters, E. H. and Bak, R. P. (1993). Effects of coral bleaching on tissue regeneration potential and colony survival. *Marine Ecology Progress Series* 189–198.

Meesters, E. H. and Bak, R. P. (1995). Age-related deterioration of a physiological function in the branching coral Acropora palmata. *Marine Ecology Progress Series* **121**, 203–209.

Meesters, E. H., Noordeloos, M. and Bak, R. P. (1994). Damage and regeneration: links to growth in the reef-building coral Montastrea annularis. *Marine Ecology Progress Series* 119–128.

Meesters, E. H., Wesseling, I. and Bak, R. P. (1996). Partial mortality in three species of reef-building corals and the relation with colony morphology. *Bulletin of Marine Science* **58**, 838–852.

Meesters, E. H., Pauchli, W. and Bak, R. P. (1997a). Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. *Marine Ecology Progress Series* **146**, 91–99.

Meesters, E. H., Wesseling, I. and Bak, R. P. (1997b). Coral colony tissue damage in six species of reef-building corals: partial mortality in relation with depth and surface area. *Journal of Sea Research* **37**, 131–144.

Meszaros, A. and Bigger, C. (1999). Qualitative and quantitative study of wound healing processes in the coelenterate, Plexaurella fusifera: spatial, temporal, and environmental (light attenuation) influences. *Journal of invertebrate pathology* **73**, 321–331.

Mullen, K. M., Peters, E. C. and Harvell, C. D. (2004). Coral resistance to disease. In *Coral health and disease*, pp. 377–399. Springer.

Mydlarz, L. D., Holthouse, S. F., Peters, E. C. and Harvell, C. D. (2008). Cellular responses in sea fan corals: granular amoebocytes react to pathogen and climate stressors. *PloS one* **3**, e1811.

Nakanishi, N., Yuan, D., Jacobs, D. K. and Hartenstein, V. (2008). Early development, pattern, and reorganization of the planula nervous system in Aurelia (Cnidaria, Scyphozoa). *Development genes and evolution* **218**, 511–524.

Olano, C. T. and Bigger, C. H. (2000). Phagocytic activities of the gorgonian coral Swiftia exserta. *Journal of invertebrate pathology* **76**, 176–184.

Oren, U., Rinkevich, B. and Loya, Y. (1997). Oriented intra-colonial transport of 14C labeled materials during coral regeneration. *Marine Ecology Progress Series* **161**, 117–122.

Palmer, C. V., Traylor-Knowles, N. G., Willis, B. L. and Bythell, J. C. (2011). Corals Use Similar Immune Cells and Wound-Healing Processes as Those of Higher Organisms. *PLOS ONE* **6**, e23992.

Passamaneck, Y. J. and Martindale, M. Q. (2012). Cell proliferation is necessary for the regeneration of oral structures in the anthozoan cnidarian Nematostella vectensis. *BMC developmental biology* **12**, 34.

Patterson, M. J. and Landolt, M. L. (1979). Cellular reaction to injury in the anthozoan Anthopleura elegantissima. *Journal of Invertebrate Pathology* **33**, 189–196.

Pfaffl, M. W. (2001). A new mathematical model for relative quantification in real-time RT–PCR. *Nucleic Acids Research* **29**, e45–e45.

Raz-Bahat, M., Erez, J. and Rinkevich, B. (2006). In vivo light-microscopic documentation for primary calcification processes in the hermatypic coral Stylophora pistillata. *Cell and tissue research* **325**, 361–368.

Reyes-Bermudez, A. and Miller, D. J. (2009). In vitro culture of cells derived from larvae of the staghorn coral Acropora millepora. *Coral Reefs* **28**, 859–864.

Reyes-Bermudez, A., Hidaka, M. and Mikheyev, A. (2021). Transcription Profiling of Cultured Acropora digitifera Adult Cells Reveals the Existence of Ancestral Genome Regulatory Modules Underlying Pluripotency and Cell Differentiation in Cnidaria. *Genome Biology and Evolution* **13**,.

Rink, J. C. (2013). Stem cell systems and regeneration in planaria. *Dev Genes Evol* **223**, 67–84.

Rinkevich, B. (1996). Do reproduction and regeneration in damaged corals compete for energy allocation? *Marine Ecology Progress Series* **143**, 297–302.

Rinkevich, B. and Loya, Y. (1979). The reproduction of the Red Sea coral Stylophora pistillata. I. Gonads and planulae. *Marine Ecology Progress Series* 133–144.

Rinkevich, Y., Paz, G., Rinkevich, B. and Reshef, R. (2007). Systemic bud induction and retinoic acid signaling underlie whole body regeneration in the urochordate Botrylloides leachi. *PLoS biology* **5**, e71.

Rinkevich, Y., Rosner, A., Rabinowitz, C., Lapidot, Z., Moiseeva, E. and Rinkevich, B. (2010). Piwi positive cells that line the vasculature epithelium, underlie whole body regeneration in a basal chordate. *Developmental biology* **345**, 94–104.

Rodríguez, S., Croquer, A., Guzmán, H. M. and Bastidas, C. (2009). A mechanism of transmission and factors affecting coral susceptibility to Halofolliculina sp. infection. *Coral reefs* **28**, 67–77.

Rogers, C. S., Gilnack, M. and Fitz III, H. C. (1983). Monitoring of coral reefs with linear transects: a study of storm damage. *Journal of Experimental Marine Biology and Ecology* **66**, 285–300.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2001). Nubbing of coral colonies: a novel approach for the development of inland broodstocks. *Aquarium Sciences and Conservation* **3**, 183–190.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2003). The use of coral nubbins in coral reef ecotoxicology testing. *Biomolecular engineering* **20**, 401–406.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2006). Coral nubbins as source material for coral biological research: a prospectus. *Aquaculture* **259**, 444–448.

Shaish, L. and Rinkevich, B. (2009). Critical evaluation of branch polarity and apical dominance as dictators of colony astogeny in a branching coral. *PloS one* **4**, e4095.

Shaish, L., Abelson, A. and Rinkevich, B. (2006). Branch to colony trajectory in a modular organism: Pattern formation in the Indo-Pacific coral Stylophora pistillata. *Developmental Dynamics* **235**, 2111–2121.

Shibata, N., Umesono, Y., Orii, H., Sakurai, T., Watanabe, K. and Agata, K. (1999). Expression ofvasa (vas)-Related genes in germline cells and totipotent somatic stem cells of planarians. *Developmental biology* **206**, 73–87.

Shukalyuk, A. I., Golovnina, K. A., Baiborodin, S. I., Gunbin, K. V., Blinov, A. G. and Isaeva, V. V. (2007). vasa-related genes and their expression in stem cells of colonial parasitic rhizocephalan barnacle Polyascus polygenea (Arthropoda: Crustacea: Cirripedia: Rhizocephala). *Cell biology international* **31**, 97–108.

Technau, U. and Steele, R. E. (2011). Evolutionary crossroads in developmental biology: Cnidaria. *Development* **138**, 1447–1458.

Titlyanov, E. A., Titlyanova, T. V., Yakovleva, I. M., Nakano, Y. and Bhagooli, R. (2005). Regeneration of artificial injuries on scleractinian corals and coral/algal competition for newly formed substrate. *Journal of Experimental Marine Biology and Ecology* **323**, 27–42.

van der Burg, C. A., Pavasovic, A., Gilding, E. K., Pelzer, E. S., Surm, J. M., Smith, H. L., Walsh, T. P. and Prentis, P. J. (2020). The rapid regenerative response of a model sea anemone species Exaiptasia pallida is characterised by tissue plasticity and highly coordinated cell communication. *Marine Biotechnology* **22**, 285–307.

Vargas-Ángel, B., Peters, E. C., Kramarsky-Winter, E., Gilliam, D. S. and Dodge, R. E. (2007). Cellular reactions to sedimentation and temperature stress in the Caribbean coral Montastraea cavernosa. *Journal of invertebrate pathology* **95**, 140–145.

Vogg, M. C., Galliot, B. and Tsiairis, C. D. (2019). Model systems for regeneration: Hydra. *Development* **146**,.

Voss, J. D. and Richardson, L. L. (2006). Nutrient enrichment enhances black band disease progression in corals. *Coral Reefs* **25**, 569–576.

Supplementary

Table S1: Primers used for qPCR assay

Gene Symbol	Accession Number	Primers	
18S	LT631145.1	Forward Reverse	GACCTGCTAAATAGTGACGCCATCCAC CGTGCGGCCCAGAACATCTAAG
EIF4A1	XM_022951886.1	Forward Reverse	CACCAGGAATGGAACAGGACGGAG ACCGTATGCGTAGATGCCTCGGAG
Piwi1	XM_022950268.1	Forward Reverse	AGCCATCACATCAGCAAGGACCACAG CCGTTGATATTGTTGAGGCATCTGTG
Nanos-1 homologue	XM_022930859.1	Forward Reverse	GTGTAGACAGCGTGGGATCGGATC TGACGTTTCAATGCACAGTTCAATGTC
Tudor5-like	XM_022924446.1	Forward Reverse	TGGTAGGAGCTCAGTCGATGGCAG GGATAGAAGAAGCCCGCGAACATTC
Tudor7-like	XM_022930561.1	Forward Reverse	GTTGCAAGCGAGGCGGACAAAC TAGAGCTGCTGATCTCCCTACACCTGG
Boule-like isoform X2	XM_022923693.1	Forward Reverse	CTATCCTCACCCTCACCCTCACCCTAG TTTGTCGAGGATTTGGAGCTTGTGGAG
Sox2-like	XM_022945978.1	Forward Reverse	GCCCGCAGCAAAGACACCTATG CAACGGCATCATGAAGGCTTCAC
Myc1-like	XM_022936348.1	Forward Reverse	TAGACGAGGACAGTTGTTCACCTGCAC TCCACATGCAGTCTTGAATGAGTTTGG

Thesis bibliography

Alié, A., Hayashi, T., Sugimura, I., Manuel, M., Sugano, W., Mano, A., Satoh, N., Agata, K. and Funayama, N. (2015). The ancestral gene repertoire of animal stem cells. *Proceedings of the National Academy of Sciences* 112, E7093–E7100.

Avasthi, S., Srivastava, R. N., Singh, A. and Srivastava, M. (2008). Stem cell: past, present and future–a review article. *Internet Journal of Medical Update* 3, 22–31.

Baums, I. B., Miller, M. W. and Szmant, A. M. (2003). Ecology of a corallivorous gastropod, *Coralliophila abbreviata*, on two scleractinian hosts. I: Population structure of snails and corals. *Marine Biology* 142, 1083–1091.

Bode, H. R. (2003). Head regeneration in *Hydra*. *Developmental dynamics: an official publication of the American Association of Anatomists* 226, 225–236.

Bonesso, J. L., Leggat, W. and Ainsworth, T. D. (2017). Exposure to elevated sea-surface temperatures below the bleaching threshold impairs coral recovery and regeneration following injury. *PeerJ* 5, e3719.

Bosch, T. C. G. (2009). *Hydra* and the evolution of stem cells. *BioEssays* 31, 478–486.

Brander, L. M., Van Beukering, P. and Cesar, H. S. (2007). The recreational value of coral reefs: a meta-analysis. *Ecological Economics* 63, 209–218.

Cesar, H., Burke, L. and Pet-Soede, L. (2003). The economics of worldwide coral reef degradation.

Cole, A. J., Pratchett, M. S. and Jones, G. P. (2008). Diversity and functional importance of coral-feeding fishes on tropical coral reefs. *Fish and Fisheries* 9, 286–307.

Counsell, C. W., Johnston, E. C. and Sale, T. L. (2019). Colony size and depth affect wound repair in a branching coral. *Marine Biology* 166, 1–12.

Cox, D. N., Chao, A., Baker, J., Chang, L., Qiao, D. and Lin, H. (1998). A novel class of evolutionarily conserved genes defined by piwi are essential for stem cell self-renewal. *Genes & development* 12, 3715–3727.

Diaz-Pulido, G., McCook, L. J., Dove, S., Berkelmans, R., Roff, G., Kline, D. I., Weeks, S., Evans, R. D., Williamson, D. H. and Hoegh-Guldberg, O. (2009). Doom and boom on a resilient reef: climate change, algal overgrowth and coral recovery. *PloS one* 4,.

Dollar, S. J. and Tribble, G. W. (1993). Recurrent storm disturbance and recovery: a long-term study of coral communities in Hawaii. *Coral Reefs* 12, 223–233.

Eh, M., W, P. and Rpm, B. (1997). Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. *Marine Ecology Progress Series* 146, 91–99.

Fujita, S., Kuranaga, E. and Nakajima, Y. (2021). Regeneration Potential of Jellyfish: Cellular Mechanisms and Molecular Insights. *Genes* 12, 758.

Funayama, N., Nakatsukasa, M., Mohri, K., Masuda, Y. and Agata, K. (2010). Piwi expression in archeocytes and choanocytes in demosponges: insights into the stem cell system in demosponges. *Evolution & development* 12, 275–287.

Gierer, A., Berking, S., Bode, H., David, C. N., Flick, K., Hansmann, G., Schaller, H. and Trenkner, E. (1972). Regeneration of *hydra* from reaggregated cells. *Nature New Biology* 239, 98–101.

Girard, A., Sachidanandam, R., Hannon, G. J. and Carmell, M. A. (2006). A germline-specific class of small RNAs binds mammalian Piwi proteins. *Nature* 442, 199–202.

Gold, D. A. and Jacobs, D. K. (2013). Stem cell dynamics in Cnidaria: are there unifying principles? *Development genes and evolution* 223, 53–66.

Gonzalez, J., Qi, H., Liu, N. and Lin, H. (2015). Piwi is a key regulator of both somatic and germline stem cells in the *Drosophila* testis. *Cell Rep* 12, 150–161.

Graham, N. A. J. and Nash, K. L. (2013). The importance of structural complexity in coral reef ecosystems. *Coral Reefs* 32, 315–326.

Grandcourt, E. M. and Cesar, H. S. (2003). The bio-economic impact of mass coral mortality on the coastal reef fisheries of the Seychelles. *Fisheries Research* 60, 539–550.

Guerrini, G., Shefy, D., Shashar, N., Shafir, S. and Rinkevich, B. (2021). Morphometric and allometric rules of polyp's landscape in regular and chimeric coral colonies of the branching species *Stylophora pistillata*. *Developmental Dynamics* 250, 652–668.

Guo, T., Peters, A. H. F. M. and Newmark, P. A. (2006). A bruno-like Gene Is Required for Stem Cell Maintenance in Planarians. *Developmental Cell* 11, 159–169.

Hall, V. R. (1997). Interspecific differences in the regeneration of artificial injuries on scleractinian corals. *Journal of Experimental Marine Biology and Ecology* 212, 9–23.

Harmelin-Vivien, M. L. (1994). The effects of storms and cyclones on coral reefs: a review. *Journal of Coastal Research* 211–231.

Henry, L.-A. and Hart, M. (2005). Regeneration from injury and resource allocation in sponges and corals–a review. *International Review of Hydrobiology: A Journal Covering all Aspects of Limnology and Marine Biology* 90, 125–158.

Horricks, R. A., Herbinger, C. M., Lillie, B. N., Taylor, P. and Lumsden, J. S. (2019). Differential protein abundance during the first month of regeneration of the Caribbean star coral *Montastraea cavernosa*. *Coral Reefs* 38, 45–61.

Horricks, R. A., Herbinger, C. M., Vickaryous, M. K., Taylor, P. and Lumsden, J. S. (2020). Differential protein abundance associated with delayed regeneration of the scleractinian coral *Montastraea cavernosa*. *Coral Reefs* 39, 1175–1186.

Hutton, D. M. and Smith, V. J. (1996). Antibacterial properties of isolated amoebocytes from the sea anemone *Actinia equina*. *The Biological Bulletin* 191, 441–451.

Ianez, R. C., Coutinho-Camillo, C. M., Buim, M. E., Pinto, C. A., Soares, F. A. and Lourenço, S. V. (2013). CD 24 and CD 44 in salivary gland pleomorphic adenoma and in human salivary gland morphogenesis: differential markers of glandular structure or stem cell indicators? *Histopathology* 62, 1075–1082.

Juliano, C. E., Reich, A., Liu, N., Götzfried, J., Zhong, M., Uman, S., Reenan, R. A., Wessel, G. M., Steele, R. E. and Lin, H. (2014). PIWI proteins and PIWI-interacting RNAs function in *Hydra* somatic stem cells. *Proceedings of the National Academy of Sciences* 111, 337–342.

Kaufman, M. L., Watkins, E., van Hooidonk, R., Baker, A. C. and Lirman, D. (2021). Thermal history influences lesion recovery of the threatened Caribbean staghorn coral *Acropora cervicornis* under heat stress. *Coral Reefs* 40, 289–293.

Lesh-Laurie, G. E., Hujer, A. and Suchy, P. (1991). Polyp regeneration from isolated tentacles of *Aurelia scyphistomae*: a role for gating mechanisms and cell division. In *Coelenterate Biology: Recent Research on Cnidaria and Ctenophora*, pp. 91–97. Springer.

Lirman, D. (2000). Lesion regeneration in the branching coral *Acropora palmata*: effects of colonization, colony size, lesion size, and lesion shape. *Marine Ecology Progress Series* 197, 209–215.

Loya, Y. (1976). The Red Sea coral *Stylophora pistillata* is an r strategist. *Nature* 259, 478–480.

Luz, B. L. P., Capel, K. C. C., Zilberberg, C., Flores, A. A. V., Migotto, A. E. and Kitahara, M. V. (2018). A polyp from nothing: The extreme regeneration capacity of the Atlantic invasive sun corals *Tubastraea coccinea* and *T. tagusensis* (Anthozoa, Scleractinia). *Journal of experimental marine biology and ecology* 503, 60–65.

Luz, B. L. P., Miller, D. J. and Kitahara, M. V. (2021). High regenerative capacity is a general feature within colonial dendrophylliid corals (Anthozoa, Scleractinia). *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* 336, 281–292.

Manikandan, B., Padelkar, A. A., Ravindran, J. and Joseph, S. (2021). Histopathological investigation of the reef coral *Goniastrea sp.* affected by macroalgal abrasion. *Marine Biology* 168, 1–7.

Mannelli, G., Magnelli, L., Deganello, A., Busoni, M., Meccariello, G., Parrinello, G. and Gallo, O. (2015). Detection of putative stem cell markers, CD 44/CD 133, in primary and lymph node metastases in head and neck squamous cell carcinomas. A preliminary immunohistochemical and in vitro study. *Clinical Otolaryngology* 40, 312–320.

Meesters, E. H. and Bak, R. P. (1993). Effects of coral bleaching on tissue regeneration potential and colony survival. *Marine Ecology Progress Series* 189–198.

Meesters, E. H. and Bak, R. P. (1995). Age-related deterioration of a physiological function in the branching coral *Acropora palmata*. *Marine Ecology Progress Series* 121, 203–209.

Meesters, E. H., Noordeloos, M. and Bak, R. P. (1994). Damage and regeneration: links to growth in the reef-building coral *Montastrea annularis*. *Marine Ecology Progress Series* 119–128.

Meesters, E. H., Wesseling, I. and Bak, R. P. (1996). Partial mortality in three species of reef-building corals and the relation with colony morphology. *Bulletin of Marine Science* 58, 838–852.

Meesters, E. H., Pauchli, W. and Bak, R. P. (1997a). Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. *Marine Ecology Progress Series* 146, 91–99.

Meesters, E. H., Wesseling, I. and Bak, R. P. (1997b). Coral colony tissue damage in six species of reef-building corals: partial mortality in relation with depth and surface area. *Journal of Sea Research* 37, 131–144.

Meszaros, A. and Bigger, C. (1999). Qualitative and quantitative study of wound healing processes in the coelenterate, *Plexaurella fusifera*: spatial, temporal, and environmental (light attenuation) influences. *Journal of invertebrate pathology* 73, 321–331.

Mullen, K. M., Peters, E. C. and Harvell, C. D. (2004). Coral resistance to disease. In *Coral health and disease*, pp. 377–399. Springer.

Muscatine, L. and Cernichiari, E. (1969). Assimilation of photosynthetic products of zooxanthellae by a reef coral. *The Biological Bulletin* 137, 506–523.

Mydlarz, L. D., Holthouse, S. F., Peters, E. C. and Harvell, C. D. (2008). Cellular responses in sea fan corals: granular amoebocytes react to pathogen and climate stressors. *PloS one* 3, e1811.

Nakanishi, N., Yuan, D., Jacobs, D. K. and Hartenstein, V. (2008). Early development, pattern, and reorganization of the planula nervous system in *Aurelia* (Cnidaria, Scyphozoa). *Development genes and evolution* 218, 511–524.

Okudela, K., Woo, T., Mitsui, H., Tajiri, M., Masuda, M. and Ohashi, K. (2012). Expression of the potential cancer stem cell markers, CD 133, CD 44, ALDH 1, and β-catenin, in primary lung adenocarcinoma—their prognostic significance. *Pathology international* 62, 792–801.

Olano, C. T. and Bigger, C. H. (2000). Phagocytic activities of the gorgonian coral *Swiftia exserta*. *Journal of invertebrate pathology* 76, 176–184.

Oren, U., Rinkevich, B. and Loya, Y. (1997). Oriented intra-colonial transport of 14C labeled materials during coral regeneration. *Marine Ecology Progress Series* 161, 117–122.

Palmer, C. V., Traylor-Knowles, N. G., Willis, B. L. and Bythell, J. C. (2011). Corals Use Similar Immune Cells and Wound-Healing Processes as Those of Higher Organisms. *PLOS ONE* 6, e23992.

Patterson, M. J. and Landolt, M. L. (1979). Cellular reaction to injury in the anthozoan *Anthopleura elegantissima*. *Journal of Invertebrate Pathology* 33, 189–196.

Raz-Bahat, M., Erez, J. and Rinkevich, B. (2006). In vivo light-microscopic documentation for primary calcification processes in the hermatypic coral *Stylophora pistillata*. *Cell and tissue research* 325, 361–368.

Rinkevich, B. (1996). Do reproduction and regeneration in damaged corals compete for energy allocation? *Marine Ecology Progress Series* 143, 297–302.

Rinkevich, Y., Rosner, A., Rabinowitz, C., Lapidot, Z., Moiseeva, E. and Rinkevich, B. (2010). Piwi positive cells that line the vasculature epithelium, underlie whole body regeneration in a basal chordate. *Developmental biology* 345, 94–104.

Rodríguez, S., Croquer, A., Guzmán, H. M. and Bastidas, C. (2009). A mechanism of transmission and factors affecting coral susceptibility to *Halofolliculina sp.* infection. *Coral reefs* 28, 67–77.

Rogers, C. S., Gilnack, M. and Fitz III, H. C. (1983). Monitoring of coral reefs with linear transects: a study of storm damage. *Journal of Experimental Marine Biology and Ecology* 66, 285–300.

Ruppert, E. E., Barnes, R. D. and Fox, R. S. (2004). *Invertebrate zoology: a functional evolutionary approach*.

Sekiné, K., Furusawa, T. and Hatakeyama, M. (2015). The boule gene is essential for spermatogenesis of haploid insect male. *Developmental biology* 399, 154–163.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2001). Nubbing of coral colonies: a novel approach for the development of inland broodstocks. *Aquarium Sciences and Conservation* 3, 183–190.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2003). The use of coral nubbins in coral reef ecotoxicology testing. *Biomolecular engineering* 20, 401–406.

Shafir, S., Van Rijn, J. and Rinkevich, B. (2006). Coral nubbins as source material for coral biological research: a prospectus. *Aquaculture* 259, 444–448.

Shaish, L. and Rinkevich, B. (2009). Critical evaluation of branch polarity and apical dominance as dictators of colony astogeny in a branching coral. *PloS one* 4, e4095.

Shaish, L., Abelson, A. and Rinkevich, B. (2006). Branch to colony trajectory in a modular organism: Pattern formation in the Indo-Pacific coral *Stylophora pistillata*. *Developmental Dynamics* 235, 2111–2121.

Shibata, N., Umesono, Y., Orii, H., Sakurai, T., Watanabe, K. and Agata, K. (1999). Expression ofvasa (vas)-Related genes in germline cells and totipotent somatic stem cells of planarians. *Developmental biology* 206, 73–87.

Shukalyuk, A. I., Golovnina, K. A., Baiborodin, S. I., Gunbin, K. V., Blinov, A. G. and Isaeva, V. V. (2007). vasa-related genes and their expression in stem cells of colonial parasitic rhizocephalan barnacle *Polyascus polygenea* (Arthropoda: Crustacea: Cirripedia: Rhizocephala). *Cell biology international* 31, 97–108.

Technau, U. and Steele, R. E. (2011). Evolutionary crossroads in developmental biology: Cnidaria. *Development* 138, 1447–1458.

Titlyanov, E. A., Titlyanova, T. V., Yakovleva, I. M., Nakano, Y. and Bhagooli, R. (2005). Regeneration of artificial injuries on scleractinian corals and coral/algal competition for newly formed substrate. *Journal of Experimental Marine Biology and Ecology* 323, 27–42.

Vargas-Ángel, B., Peters, E. C., Kramarsky-Winter, E., Gilliam, D. S. and Dodge, R. E. (2007). Cellular reactions to sedimentation and temperature stress in the Caribbean coral *Montastraea cavernosa*. *Journal of invertebrate pathology* 95, 140–145.

Vaucheret, H. (2008). Plant ARGONAUTES. *Trends in Plant Science* 13, 350–358.

Veron, J. E. N. and Stafford-Smith, M. (2000). *Corals of the world*. Townsville MC, Qld, Australia: Australian Institute of Marine Science.

Vogg, M. C., Galliot, B. and Tsiairis, C. D. (2019). Model systems for regeneration: *Hydra*. *Development* 146,.

Voss, J. D. and Richardson, L. L. (2006). Nutrient enrichment enhances black band disease progression in corals. *Coral Reefs* 25, 569–576.

Zhao, W., Ji, X., Zhang, F., Li, L. and Ma, L. (2012). Embryonic stem cell markers. *Molecules* 17, 6196–6236.

Supplementary

Table S1: Primers used for qPCR assay.

Gene Symbol	Accession Number	Primers	
18S	LT631145.1	Forward Reverse	GACCTGCTAAATAGTGACGCCATCCAC CGTGCGGCCCAGAACATCTAAG
EIF4A1	XM_022951886.1	Forward Reverse	CACCAGGAATGGAACAGGACGGAG ACCGTATGCGTAGATGCCTCGGAG
Piwi1	XM_022950268.1	Forward Reverse	AGCCATCACATCAGCAAGGACCACAG CCGTTGATATTGTTGAGGCATCTGTG
Nanos-1 homologue	XM_022930859.1	Forward Reverse	GTGTAGACAGCGTGGGATCGGATC TGACGTTTCAATGCACAGTTCAATGTC
Tudor5-like	XM_022924446.1	Forward Reverse	TGGTAGGAGCTCAGTCGATGGCAG GGATAGAAGAAGCCCGCGAACATTC
Tudor7-like	XM_022930561.1	Forward Reverse	GTTGCAAGCGAGGCGGACAAAC TAGAGCTGCTGATCTCCCTACACCTGG
Boule-like isoform X2	XM_022923693.1	Forward Reverse	CTATCCTCACCCTCACCCTCACCCTAG TTTGTCGAGGATTTGGAGCTTGTGGAG
Sox2-like	XM_022945978.1	Forward Reverse	GCCCGCAGCAAAGACACCTATG CAACGGCATCATGAAGGCTTCAC
Myc1-like	XM_022936348.1	Forward Reverse	TAGACGAGGACAGTTGTTCACCTGCAC TCCACATGCAGTCTTGAATGAGTTTGG

Table S2: Injury surface area [mm^2] for each nubbin at each day.

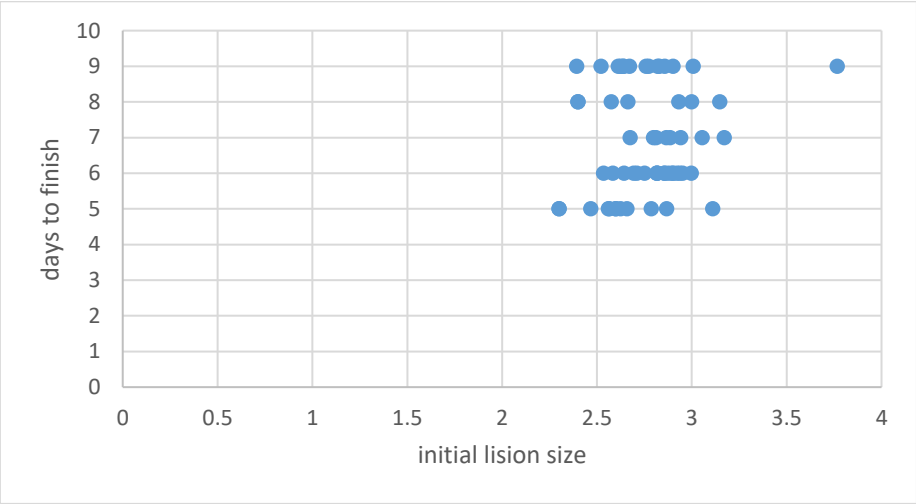
Genet #	Nubbin #	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
1	01	2.3	1.6	1	0.4	0.1	0	0	0	0	0
1	02	2.6	1.9	1.7	0.6	0.2	0	0	0	0	0
1	03	2.9	2.2	1.6	1.5	0.7	0.3	0	0	0	0
1	04	2.4	1.7	1.5	1.1	0.4	0.5	0.4	0.1	0	0
1	3	2.59	2.589	2.277	1.932	2.002	1.674	1.444	1.094	0.792	0.234
1	4	2.393	2.103	2.067	2.083	1.996	1.344	0.821	0.272	0.005	0
1	5	2.798	2.482	2.567	2.083	1.219	0.539	0.196	0	0	0
1	6	2.613	1.91	2.045	1.563	1.119	0.596	0.308	0.1	0.075	0
1	8	2.672	2.35	2.865	2.545	2.112	1.35	0.272	0.171	0.08	0
1	9	2.521	2.245	1.846	1.458	1.115	0.807	0.342	0.143	0.02	0
1	10	3.466	2.776	2.07	1.403	1.183	1.012	0.813	0.632	0.384	0.069
1	12	2.644	2.448	2.343	2.03	1.14	0.01	0	0	0	0
1	13	2.535	2.283	1.764	0.698	0.132	0.873	0	0	0	0
1	14	2.931	2.429	2.172	1.545	1.484	0.783	0.614	0.221	0	0
1	15	2.3	1.6	1	0.4	0.1	0	0	0	0	0
1	16	2.6	1.9	1.7	0.6	0.01959	0	0	0	0	0
1	17	2.9	2.2	1.6	1.5	0.7	0.3	0	0	0	0
1	18	2.4	1.7	1.5	1.1	0.4	0.5	0.4	0.1	0	0
2	1	2.658	2.081	1.412	0.669	0.204	0	0	0	0	0
2	2	2.468	1.948	1.467	0.626	0.209	0	0	0	0	0
2	3	2.786	2.435	1.615	0.598	0.1	0	0	0	0	0
2	5	2.71	2.267	2.919	1.434	0.821	0.31	0	0	0	0
2	6	3.008	3.073	2.538	2.43	1.985	1.278	0.661	0.917	0.889	0
2	8	2.624	2.345	1.651	1.663	1.447	0.907	0.5	0.276	0.127	0
2	9	2.693	2.355	1.473	0.86	0.342	0	0	0	0	0
2	10	2.856	2.173	1.221	0.604	0.034	0	0	0	0	0
2	11	2.997	2.458	1.918	0.265	0.002	0	0	0	0	0

2	12	2.817	2.319	2.116	1.099	0.371	0	0	0	0	0
2	14	2.638	2.297	1.798	1.9	1.82	1.627	1.129	0.67	0.267	0
2	15	2.818	2.23	1.867	0.942	0.316	0.011	0	0	0	0
2	16	2.625	2.22	1.573	0.897	0.335	0	0	0	0	0
2	17	2.952	2.194	1.777	1.131	0.805	0.387	0	0	0	0
2	18	2.566	2.313	1.609	0.745	0.048	0	0	0	0	0
2	19	2.584	2.317	1.658	1.195	0.837	0.107	0	0	0	0
2	20	2.649	2.224	2.151	0.952	0.443	0.171	1.103	0.979	0.611	0.022
2	21	2.561	2.249	1.839	0.95	0.137	0	0	0	0	0
2	22	2.812	2.394	1.719	1.659	1.578	1.258	0.239	0	0	0
2	23	3.11	2.501	1.53	0.503	0.144	0	0	0	0	0
2	24	2.815	2.258	2.014	0.681	0.179	0.004	0	0	0	0
2	26	2.936	2.234	3.167	1.535	1.167	0.574	0	0	0	0
2	27	2.868	2.122	2.001	1.101	0.558	0	0	0	0	0
2	29	2.877	2.412	1.992	1.087	0.765	0.271	0	0	0	0
2	31	2.866	2.548	2.058	1.145	0.385	0.096	0.008	0	0	0
2	32	2.886	2.435	1.962	1.264	0.933	0.465	0.269	0	0	0
2	33	2.752	2.203	2.07	1.288	1.164	0.67	0	0	0	0
3	1	2.722	2.565	2.53	2.194	2.196	1.776	1.385	1.028	1.165	0.512
3	2	2.799	2.693	2.767	2.55	2.635	2.747	2.314	1.7	1.27	0.374
3	3	2.911	3.115	3.169	2.851	2.384	2.173	1.692	1.528	1.386	0.963
3	4	2.54	2.349	2.381	1.865	1.425	0.956	0.739	0.522	0.376	0.046
3	5	2.396	2.693	1.905	1.633	1.336	1.411	0.991	0.876	0.522	0.226
3	6	2.631	2.403	2.502	1.944	1.916	1.822	1.318	1.005	0.753	0.356
3	8	2.575	2.278	2.228	1.916	1.649	1.18	0.741	0.381	0	0
3	9	2.857	2.325	2.147	1.253	0.53	0.019	0	0	0	0
3	10	2.676	2.421	2.469	1.572	0.909	0.332	0.006	0	0	0
3	11	2.773	2.379	2.468	2.107	1.984	1.538	0.763	0.152	0.197	0
3	12	3.767	3.733	3.38	2.832	2.397	1.923	1.265	0.864	0.528	0
3	13	2.858	2.533	2.35	1.836	1.437	1.149	0.695	0.404	0.065	0
3	14	2.977	2.997	3.008	2.375	1.907	1.386	0.665	0.286	0.218	0.116
3	15	2.999	2.776	2.78	2.209	1.994	1.092	0.685	0.41	0	0
3	16	2.663	2.762	2.919	2.444	2.303	1.618	0.751	0.13	0	0
3	17	2.942	2.704	2.575	1.703	1.211	0.526	0.103	0	0	0
3	18	2.76	2.69	2.812	2.184	1.89	1.298	0.562	0.164	0.19	0
3	19	2.901	2.874	3.128	2.737	2.811	2.388	1.836	1.411	0.984	0
3	20	2.82	3.025	3.021	2.415	2.319	1.846	1.019	0.498	0.077	0
3	21	2.824	2.515	2.61	2.201	1.789	1.312	0.801	0.887	0.858	0.183
3	22	3.172	2.637	2.292	1.581	1.06	0.415	0.019	0	0	0
3	23	3.147	2.78	2.686	1.874	1.412	0.617	0.211	0.017	0	0
3	24	2.642	3.086	2.991	2.778	2.572	2.455	1.569	1.149	0.225	0
3	25	2.829	2.489	2.405	2.057	2.193	2.152	1.469	0.75	0.374	0
3	26	3.055	2.489	2.297	1.436	0.762	0.2	0.006	0	0	0
3	27	2.921	2.398	2.327	1.721	0.933	0.229	0	0	0	0

Table S3: Statistical analyses to total regeneration rate between each genet

Test Statistics[a,b]		Test Statistics[a]		Test Statistics[a]		Test Statistics[a]	
	Rate	Genet 1-2	Rate	Genet 1-3	Rate	Genet 2-3	Rate
Kruskal-Wallis H	13.359	Mann-Whitney U	117.000	Mann-Whitney U	133.000	Mann-Whitney U	84.000
df	2	Wilcoxon W	253.000	Wilcoxon W	304.000	Wilcoxon W	255.000
Asymp. Sig.	0.001	Z	-2.357	Z	-0.380	Z	-3.581
a. Kruskal Wallis Test		Asymp. Sig. (2-tailed)	0.018	Asymp. Sig. (2-tailed)	0.704	Asymp. Sig. (2-tailed)	0.000
b. Grouping Variable: Genet		a. Grouping Variable: Genet		Exact Sig. [2*(1-tailed Sig.)]	.721[b]	a. Grouping Variable: Genet	
				a. Grouping Variable: Genet b. Not corrected for ties.			

Figure S4: Scatter plot of initial injury size (x axis) and days to full regeneration (y axis) and Spearman's test for correlation between them.



Correlations

			Injury_surface_area	Days to FR
Spearman's rho	Injury_surface_area	Correlation Coefficient	1.000	.137
		Sig. (2-tailed)	.	.297
		N	60	60
	Days to FR	Correlation Coefficient	.137	1.000
		Sig. (2-tailed)	.297	.
		N	60	60

Figure S5: Line plot of injury days to full regeneration (x axis) and surface area (y axis).

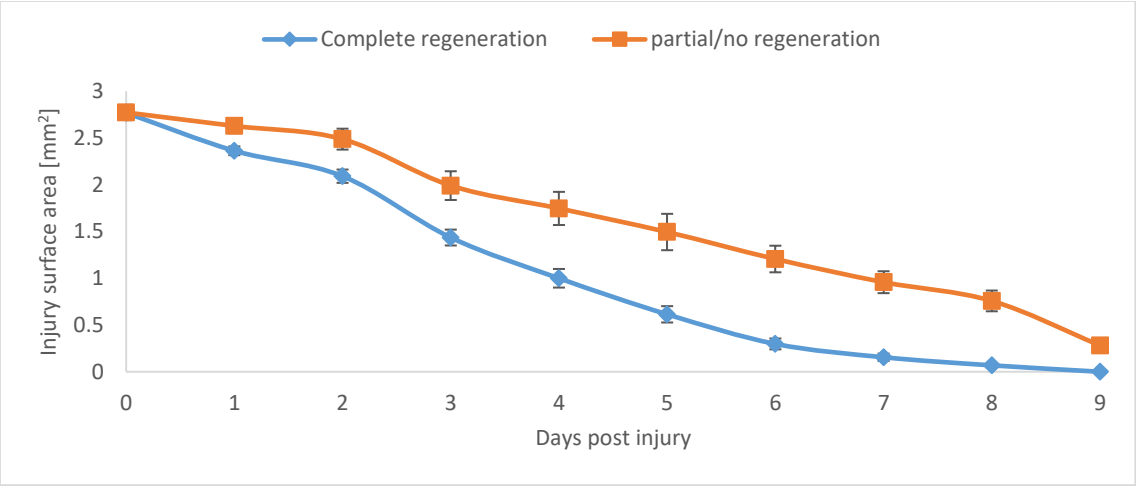


Figure S6: Chi2 test to number of polyps rebuilt following tissue regeneration.

Test Statistics

No. of polyps	Observed N	Expected N	Residual
2-3	4	21.3	-17.3
1	22	21.3	0.7
0	38	21.3	16.7
Total	64		

	No_polyp
Chi-Square	27.125[a]
df	2
Asymp. Sig.	0.000

a. 0 cells (0.0%) have expected frequencies less than 5. The minimum expected cell frequency is 21.3.

גדילת והתחדשות רקמת האלמוגים מלווה בהופעתם של תאים דמויי תאי גזע בוגרים

יונתן לבנוני

תקציר

אחת מיכולותיהם המשמעותיות והחשובות של מושבות האלמוגים הינה יכולת רגנרציה, בניה מחדש. המושבות יכולות לבצע רגנרציה של פצעים קטנים ברקמה המקשרת, פוליפים שלמים ואף יצירה מחדש של מבנה מושבה שלם. תכונות רגנרטיביות אלה מאפשרות לאלמוגים לגדל מחדש חלקים שנפגעו בעקבות שלל ומגוון רחב של פגיעות ביולוגיות ופיזיקליות שמקורן בגורמים ביולוגיים, פיזיקאליים ואנתרופוגניים. על אף שתהליך בניית הרקמה מחדש (רגנרציה) לאחר השרייה של שלל פציעות ופגיעות נחקר רבות באורגניזמים ממחלקת האלמוגים לאורך זה למעלה מ100 שנים, המנגנונים התאיים והרקמתיים הלוקחים חלק בתהליך זה אינם מוכרים דיים ועדויות לקיומם חסרות. במחקר זה, תהליך הרגנרציה נבחן לאחר ביצוע פציעות קטנות ועגולות, ששטחן כ- 2.77ממ^2 ובהן הוסר פוליפ יחיד. התהליך נבחן באמצעות צילומים מיקרוסקופיים מדי 24 שעות מרגע הפציעה ועד סגירתה, חתכים היסטולוגיים, צביעה באמצעות נוגדנים לסמן להתחלקות תאים (pH3) ולסמן לתאי גזע Piwi, וכן בחינת ביטוי גנים הקשורים בתהליכים של תאי גזע באמצעות מכשיר qPCR. לאחר כ-9 ימים מיום הפציעה, כ-84.5% (n=64) מהפציעות עברו החלמה מלאה (נסגרו לחלוטין). בכ-35% מהמקרים גדל פוליפ אחד בחזרה, בעוד בכמעט 60% לא גדל כלל פוליפ חדש ואף בכ-6% גדלו שניים או שלושה פוליפים חדשים. החתכים ההיסטולוגיים חשפו הופעה של מבנה ייחודי של התארכות האפידרמיס ברקמה הנבנית מחדש, כמו גם הגדלה ומתפשטת על המצע, המגיע עד כדי 150µm לערך. מבנה מוארך זה מכיל מקבצים של תאים רבים וקטנים, שלהם גרעין דחוס, המזכירים תאי גזע. רבים מתאים אלה נצבעו הן בנוגדנים לpH3 וPiwi. כמו כן, כל המרקרים הגנטיים לתאי גזע הראו שונות מובהקת סטטיסטית בביטויים בין המצבים הביולוגיים שנבחנו. רובם הראו עלייה מובהקת בביטוי במצב של רקמה שזה עתה סיימה את איחוי הפצע או ברקמה הנמצאת בגדילה. מחקר זה חשף לראשונה באלמוגים (Anthozoa) את הנוכחות וההשתתפות של תאים דמויי תאי גזע בוגרים בתהליך הרגנרציה והגדילה הטבעית, השוכנים בחזית הגדילה של הרקמה. אנו גורסים כי באלמוגים מתבצע שימוש במנגנון התחדשות רקמה וגדילה דומה לשאר האורגניזמים במערכת הצורבים. יתכן כי ממצא זה מחל בגישור הפער בין חברי מערכה זו בכל הקשור בקיומם של תאי גזע בוגרים ותפקידיהם.

גדילת והתחדשות רקמת האלמוגים מלווה בהופעתם של תאים דמויי תאי גזע בוגרים

מאת: יונתן לבנוני

בהנחיית: פרופ' ברוך רינקביץ'

עבודת גמר מחקרית (תזה) המוגשת כמילוי חלק מהדרישות לקבלת התואר "מוסמך האוניברסיטה"

אוניברסיטת חיפה

הפקולטה למדעי הטבע

בית הספר למדעי הים על שם ליאון צ'רני

החוג לביולוגיה ימית

נובמבר, 2021

גדילת והתחדשות רקמת האלמוגים מלווה בהופעתם של תאים דמויי תאי גזע בוגרים

יונתן לבנוני

עבודת גמר מחקרית (תזה), המוגשת כמילוי חלק מהדרישות לקבלת התואר "מוסמך האוניברסיטה"

אוניברסיטת חיפה

הפקולטה למדעי הטבע

בית הספר למדעי הים על שם ליאון צ'רני

החוג לביולוגיה ימית

נובמבר, 2021

ProQuest Number: 29133597

INFORMATION TO ALL USERS
The quality and completeness of this reproduction is dependent on the quality
and completeness of the copy made available to ProQuest.

Distributed by ProQuest LLC (2022).
Copyright of the Dissertation is held by the Author unless otherwise noted.

This work may be used in accordance with the terms of the Creative Commons license
or other rights statement, as indicated in the copyright statement or in the metadata
associated with this work. Unless otherwise specified in the copyright statement
or the metadata, all rights are reserved by the copyright holder.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

Microform Edition where available © ProQuest LLC. No reproduction or digitization
of the Microform Edition is authorized without permission of ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346 USA